Syniorage™

“Senior” anti-age targeting the epidermis

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Syniorage™ – Concept

Seniors increasingly carry wrinkles, the main consequences of dermal aging, with pride, viewing them as evidence of life's rich experiences.

The signs of epidermal aging, such as sallow, fragile and thin skin remain less accepted.

Seniors expect their skin to mirror their joy in life. They prefer to project a youthful image, reflected by a radiant complexion.

Complementary to most anti-aging actives that target the dermis, Syniorage™ targets the epidermis.

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Syniorage™ – Definition

A pure **tetrapeptide that reinforces**

The epidermal cohesion via a specific small proteoglycan: **syndecan-1**

The adhesion between epidermis and dermo-epidermal junction via a fundamental protein of the hemidesmosomes: **collagen XVII**

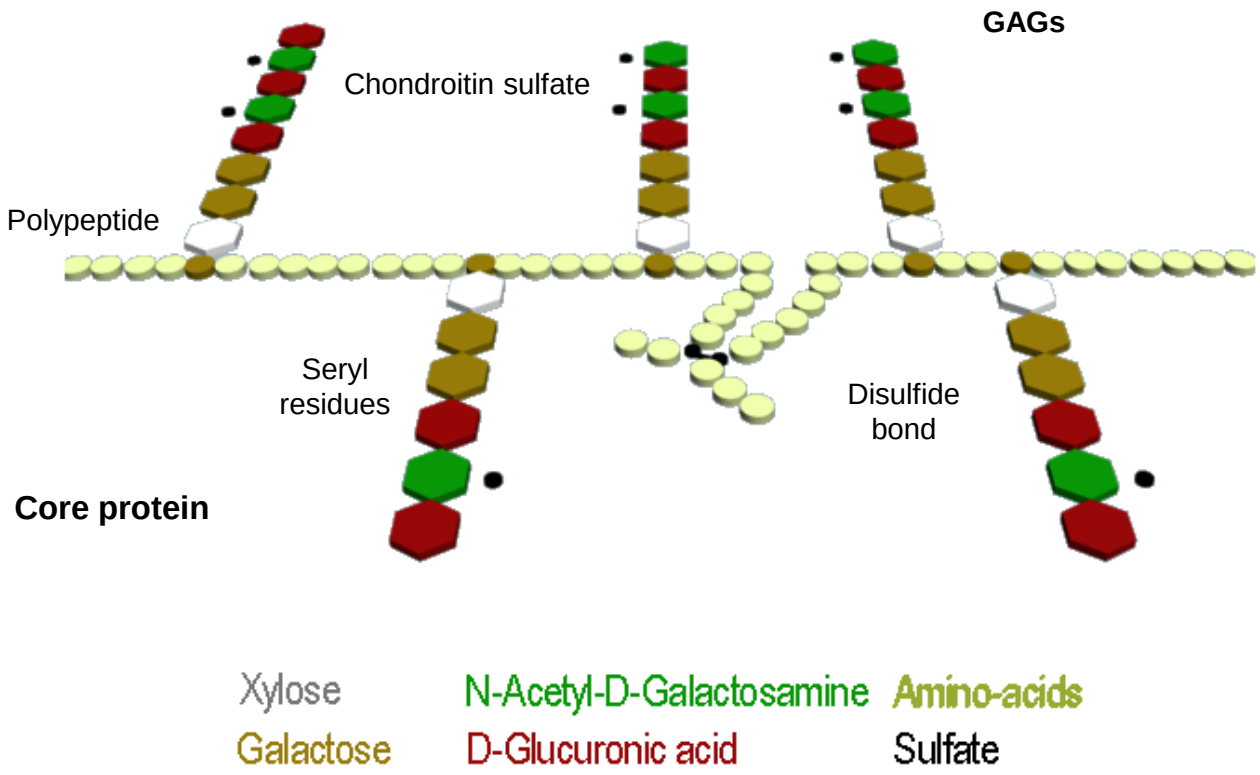
For a **firmer, more resistant skin with refined texture**

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Concept

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Proteoglycans (Pgs): Definition and Functions



⇨**Biomechanical support**

⇨**Biological filter**

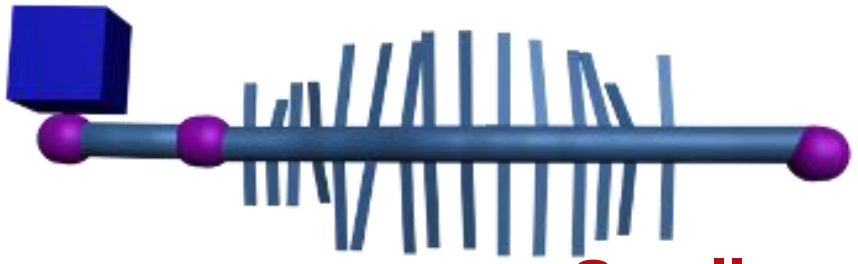
⇨**Tissue organizer**:
- proliferation
- migration
- differentiation

Homeostasis

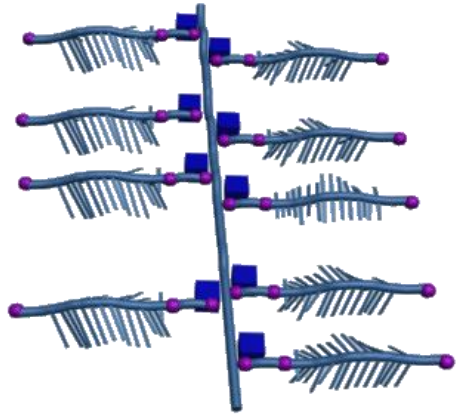
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Proteoglycans: Organization

Extracellular large proteoglycans

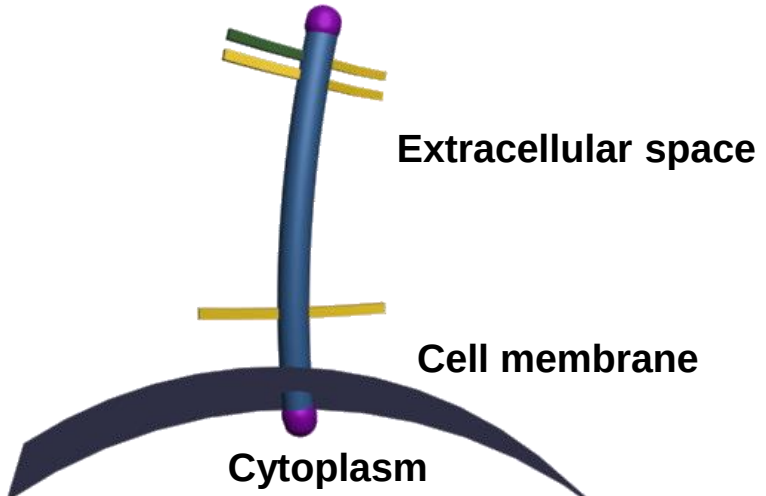


Aggregates of proteoglycans



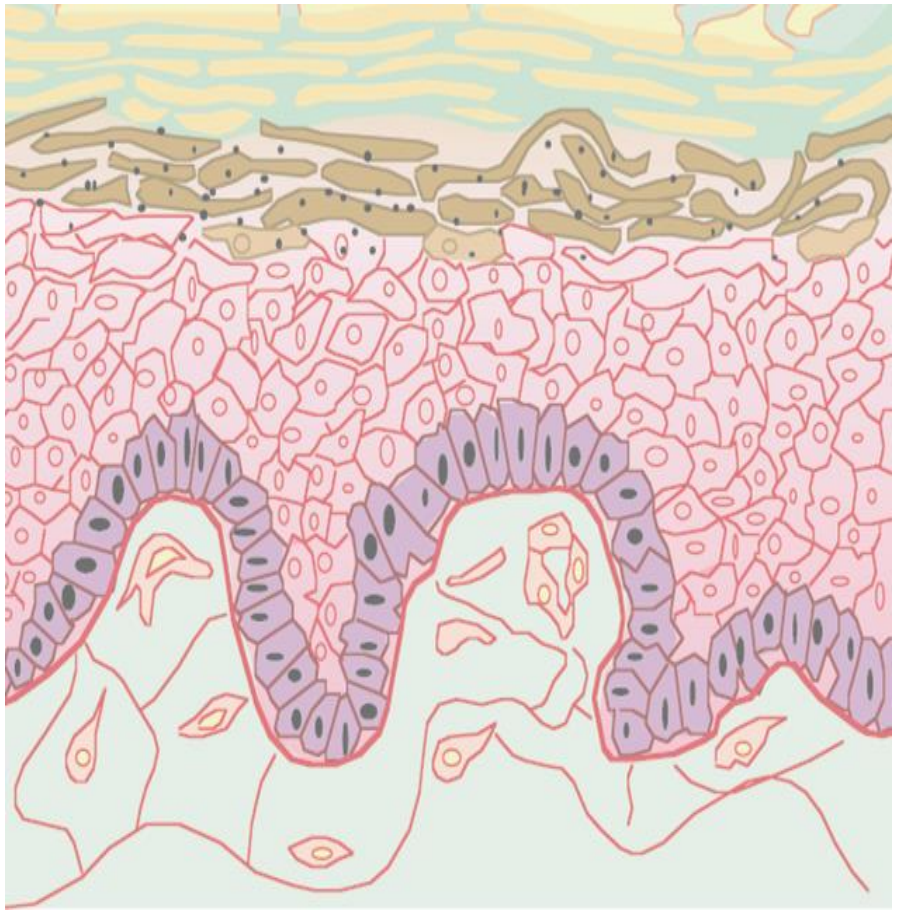
Small proteoglycans (< 60 000 Da)

Syndecan



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Syndecans: Small Pgs In The Epidermis



Syndecan-1
(HS/CS)

- Cell signaling
- Cell matrix adhesion
- Cell-cell adhesion
- Cell migration

Syndecan-4
(HS/CS)

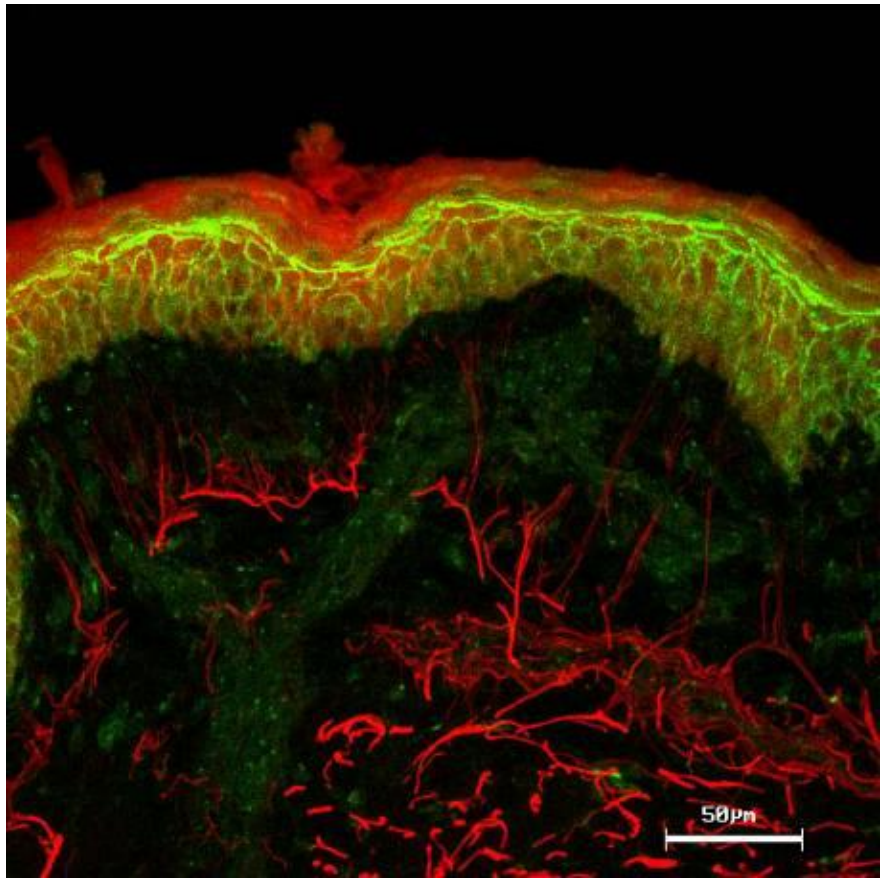
Role in epidermal cohesion

Carey DJ: Syndecans: multifunctional cell-surface co-receptors. Biochem J, 327, 1-16, 1997

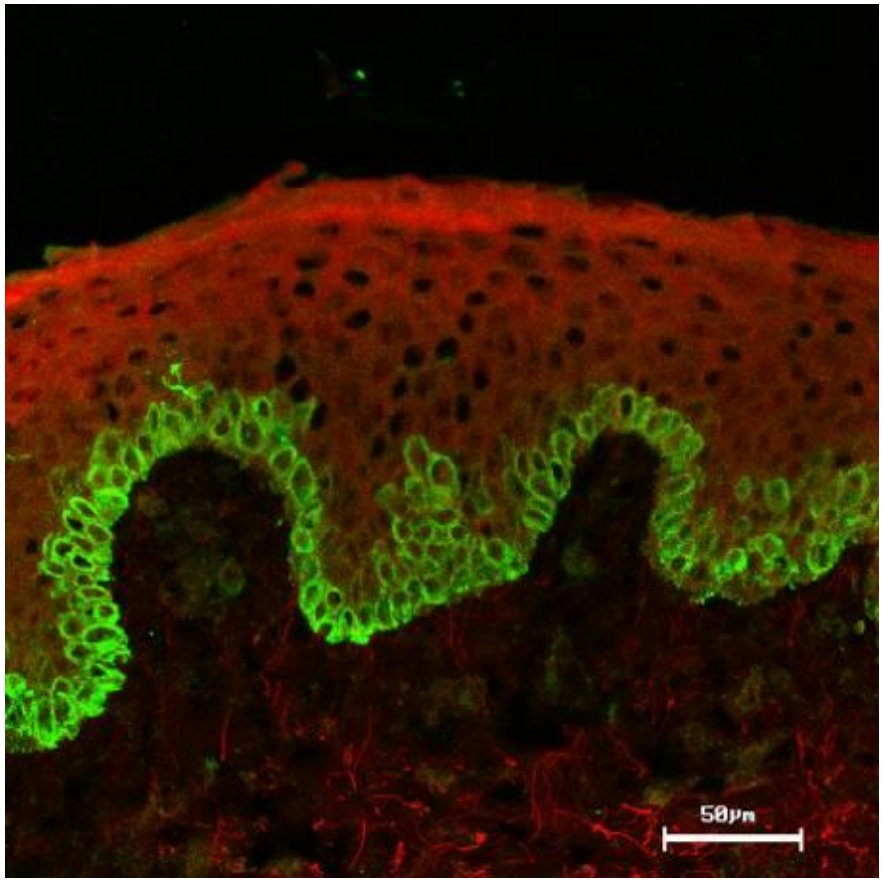
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Syndecans: visualization in human epidermis

Syndecan-1 in suprabasal layers



Syndecan-4 in basal layer



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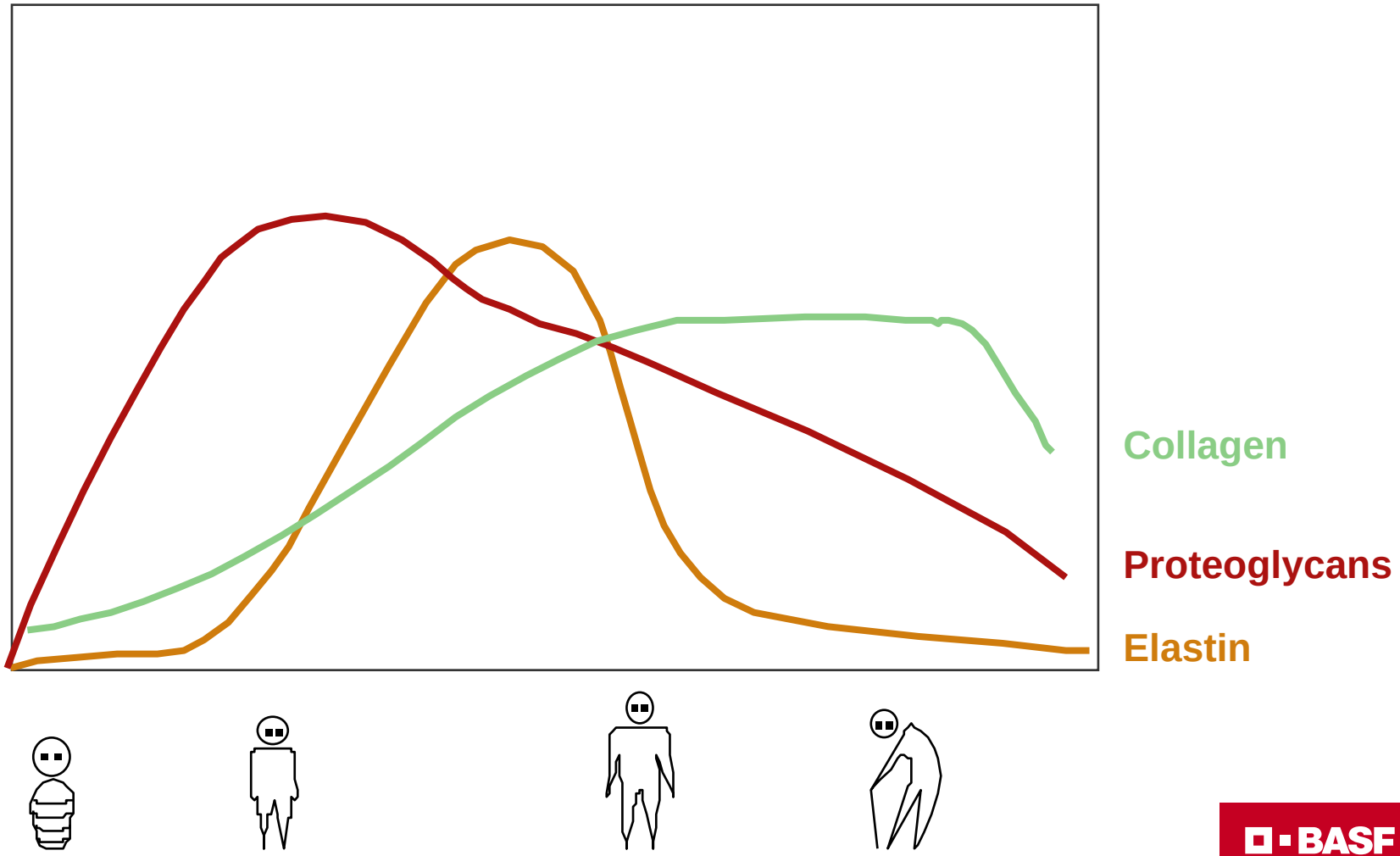
Glycosaminoglycans, Proteoglycans And Skin Aging: A Few References

- ↘ GAGs synthesis in adult skin / fetal skin (Breen, 1970)
- ↘ hyaluronic acid and dermatan sulfate in aged skin (Bernstein, 1995)
- ↘ PGs in the skin with aging (Winkler, 2001)
- ↘ large PGs/small PG in fibroblasts culture (Hayflick model, 1974)
- ↘ ratio biglycan / decorin in aged dermal fibroblasts (Schonherr, 1993)
- ↗ decorin / age (Roughley,1994 – Carino, 2000 – Del Santo, 2000)

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Glycosaminoglycans and proteoglycans: variation with age

Biosynthesis



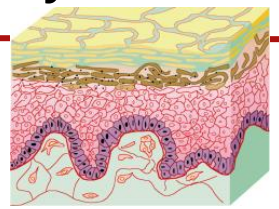
L. Robert In: Précis de physiologie cutanée, Meynadier J ed, 259-266, 1980

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Syndecan-1 during skin aging – Fundamental study

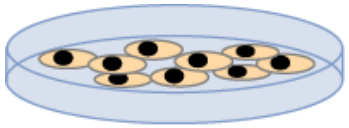
Human cutaneous biopsies

1 month to 83 years old (36 donors)



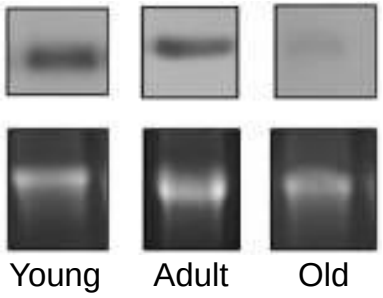
Culture

- Epidermal keratinocytes (primary culture)



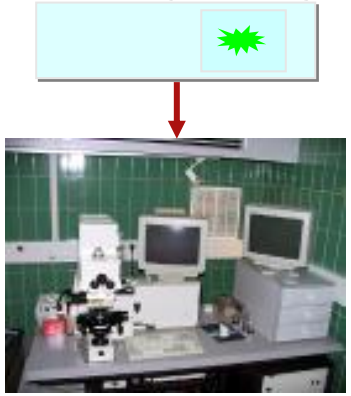
Proteoglycan mRNA level

- Northern-blot
- Quantification



Proteoglycan protein level on cutaneous biopsies

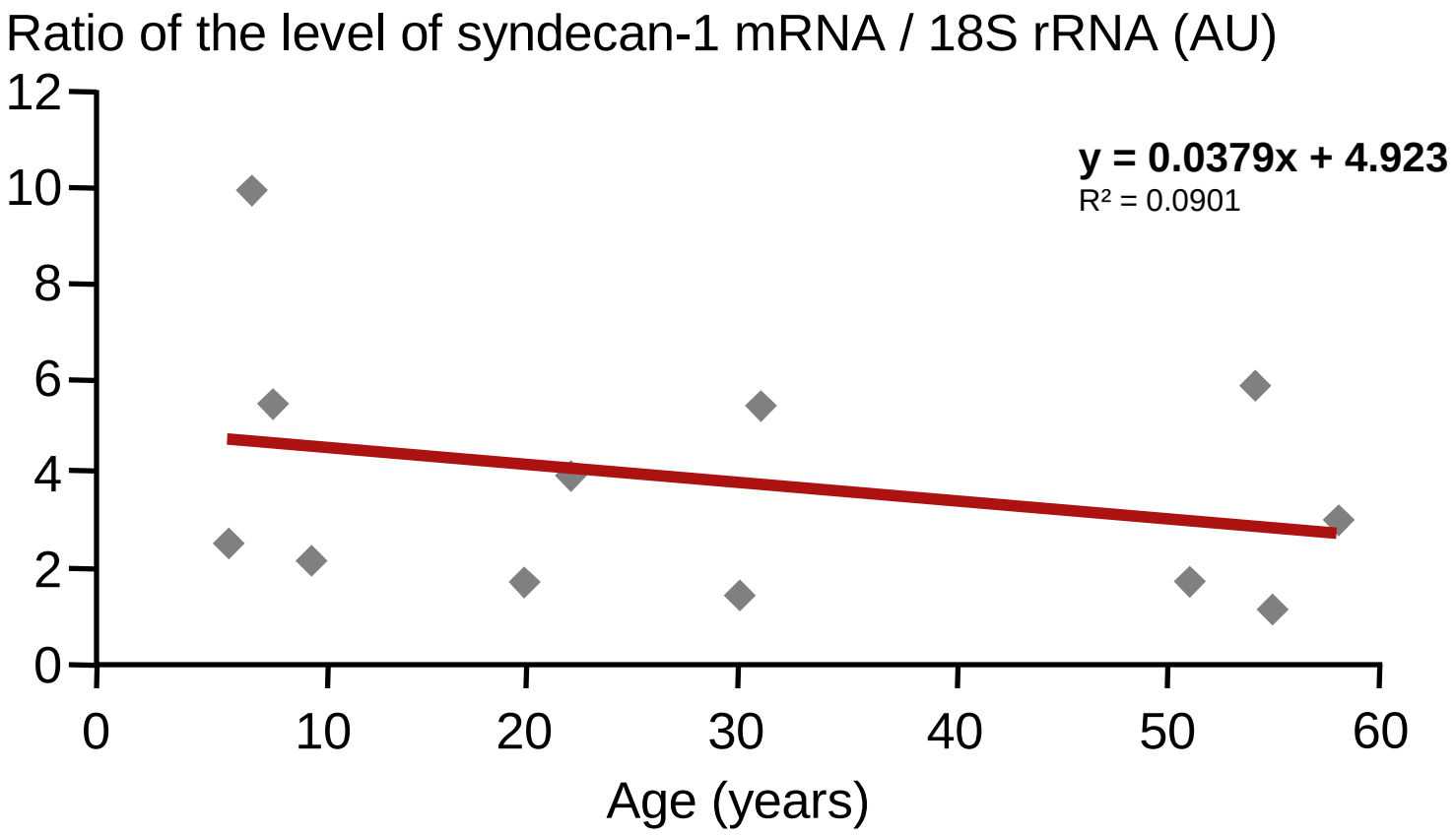
- Immunohistochemistry (IHC)
- Confocal microscopy
- Quantification by image analysis



Wegrowski Y, Danoux L, Contet-Audonneau JL, Pauly G, Maquart FX: Decreased syndecan-1 expression by human keratinocytes during skin aging. J Invest Dermatol 125: 5, Abstracts 029, 2005

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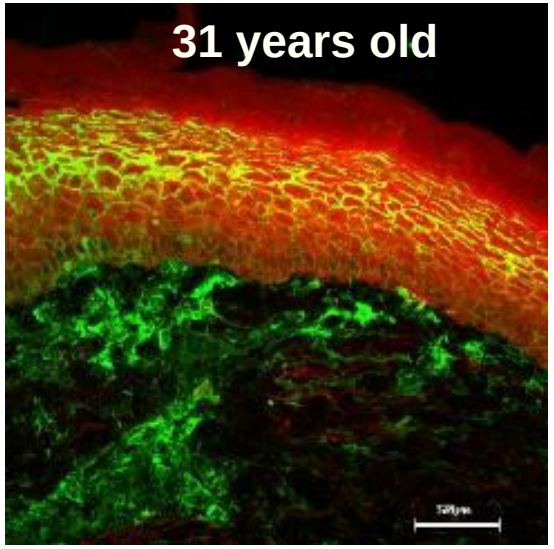
Syndecan-1 During Skin Aging – Variation In Keratinocytes

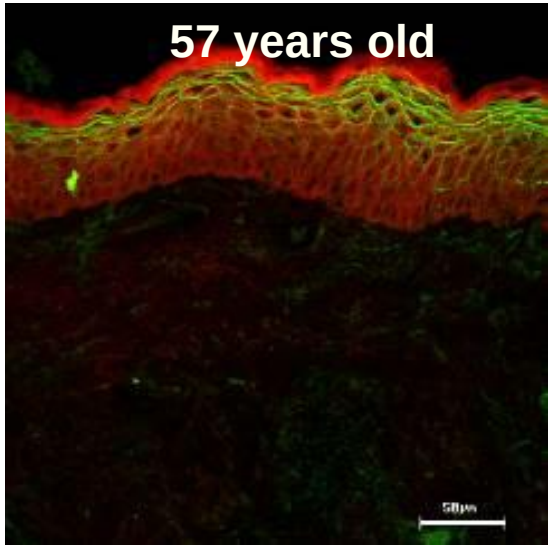


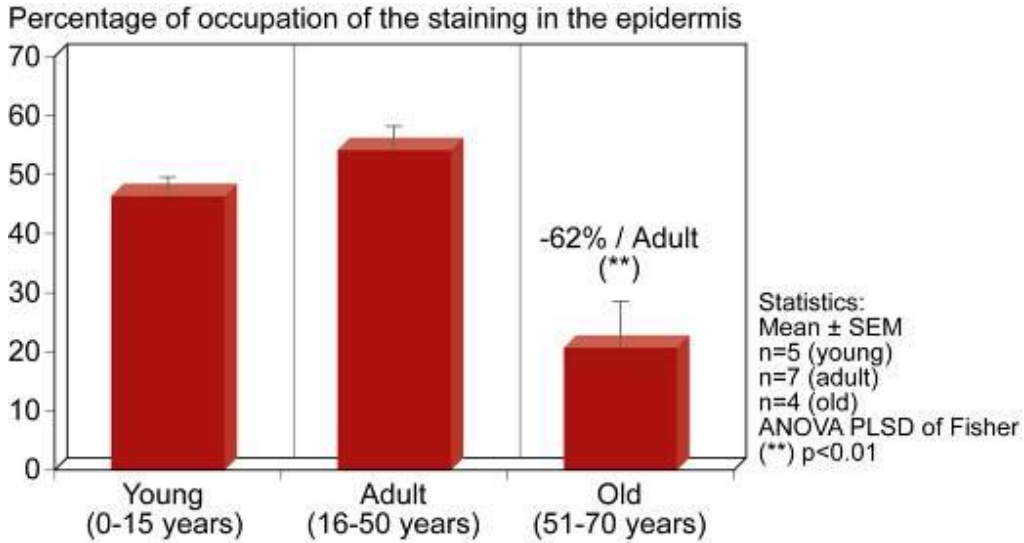
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Syndecan-1: variation in the epidermis with age (IHC – Confocal microscopy)









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Collagen XVII: Collagenous Transmembrane Protein

Definition: initially identified as the 180-kDa bullous pemphigoid antigen-2 (BP180 or BPAg2):

- homotrimer, consisting of 3 identical α1 (XVII) chains (Col17A1),
- structural component of the dermo-epidermal anchoring complex.

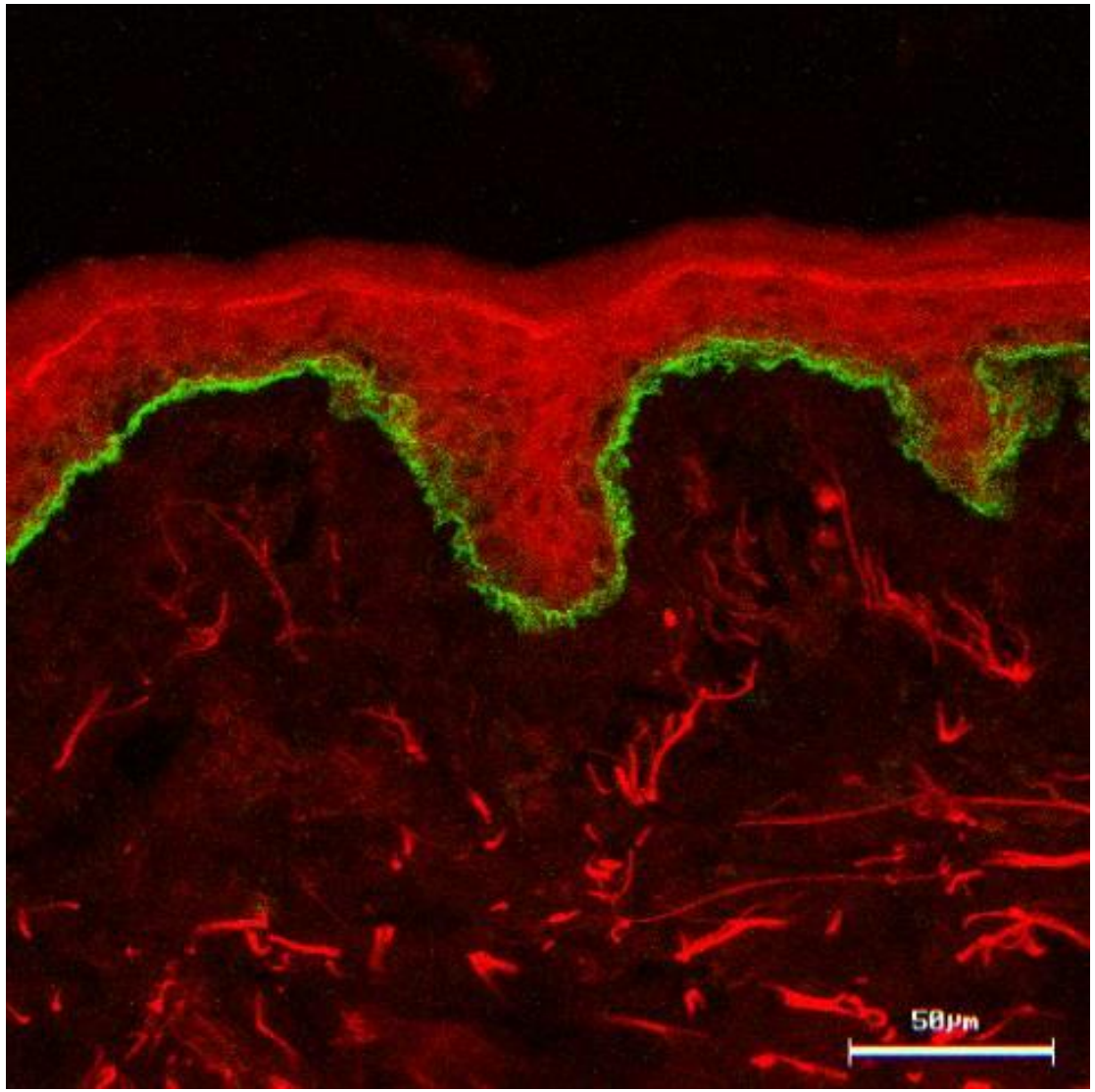
Location: Lumina lucida of dermo-epidermal junction (DEJ).

Main function: cell-matrix adhesion molecule through stabilization of the hemidesmosome complex.



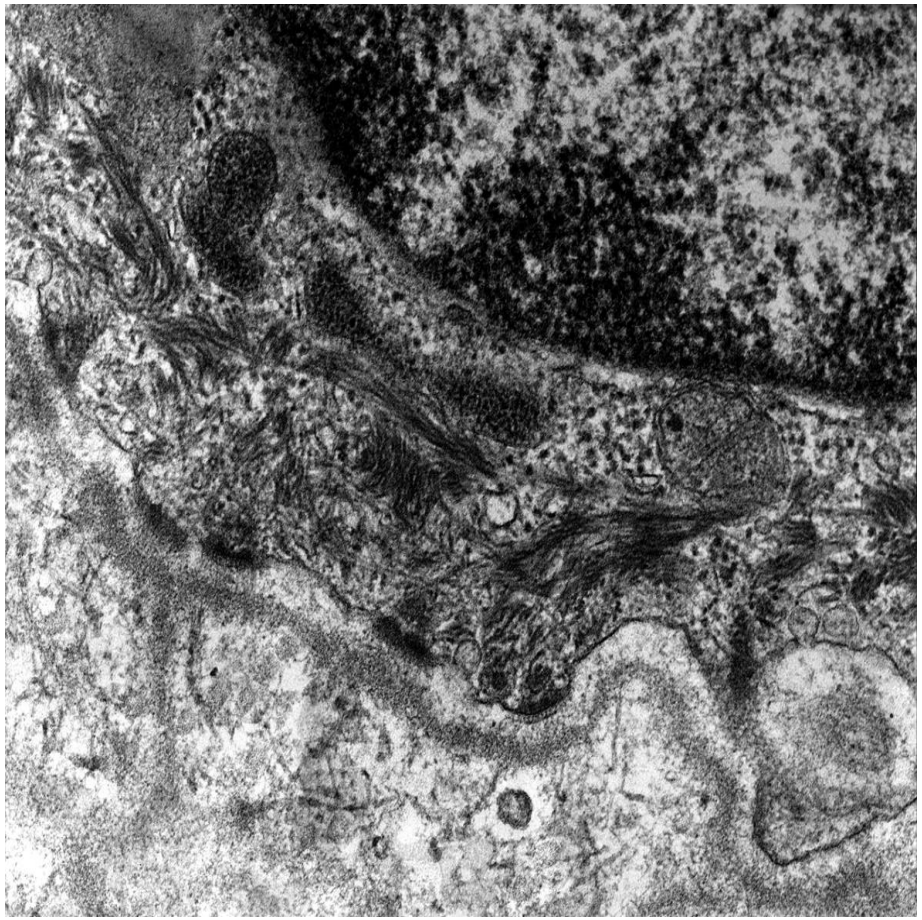
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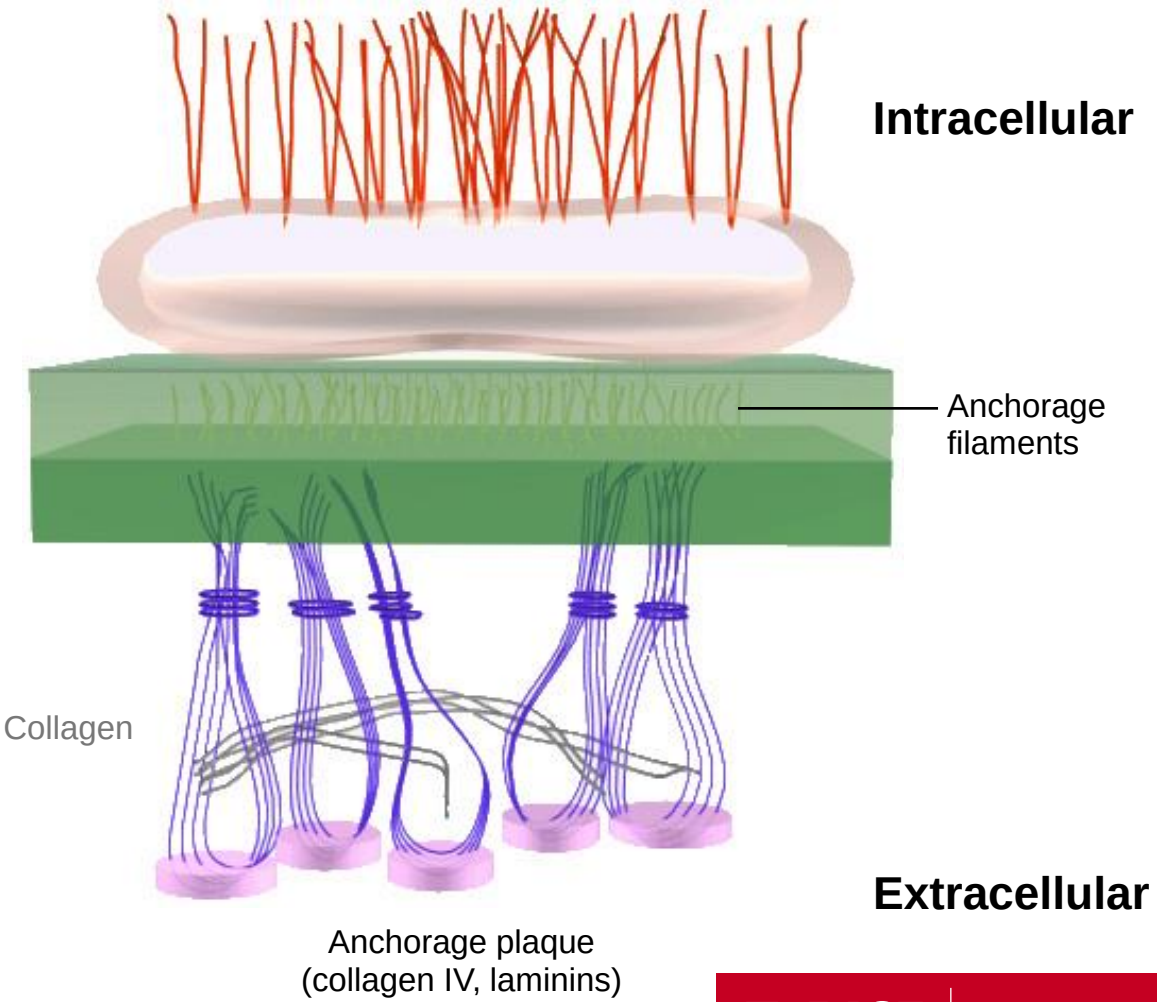
Collagen XVII: Visualization In Human Epidermis (IHC – Confocal Microscopy)



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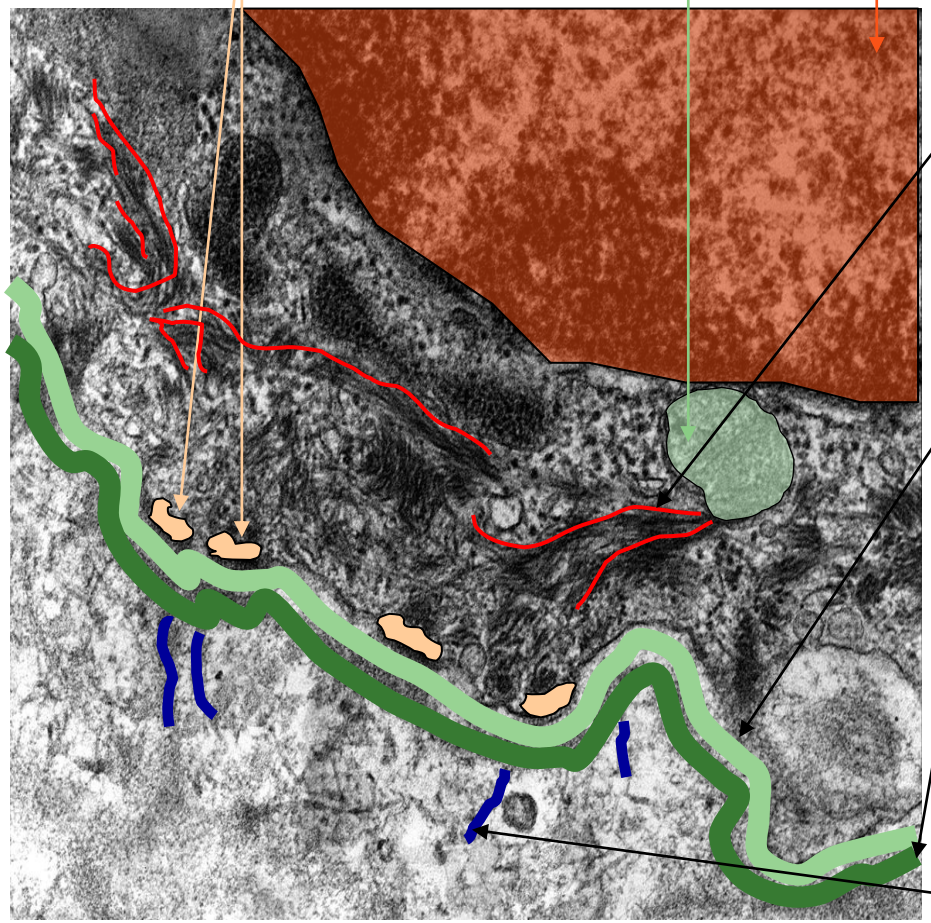
DEJ: Structural Organization

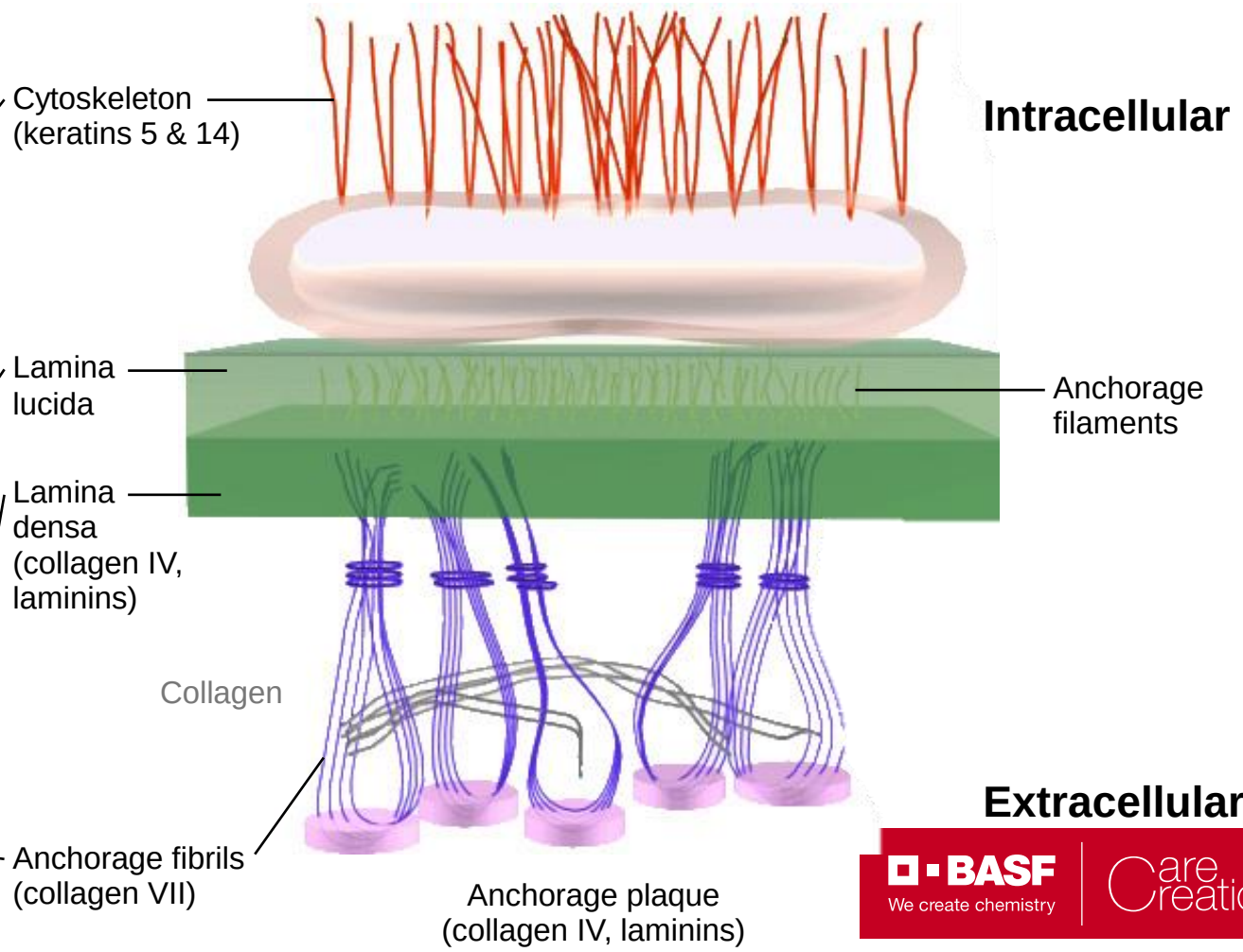




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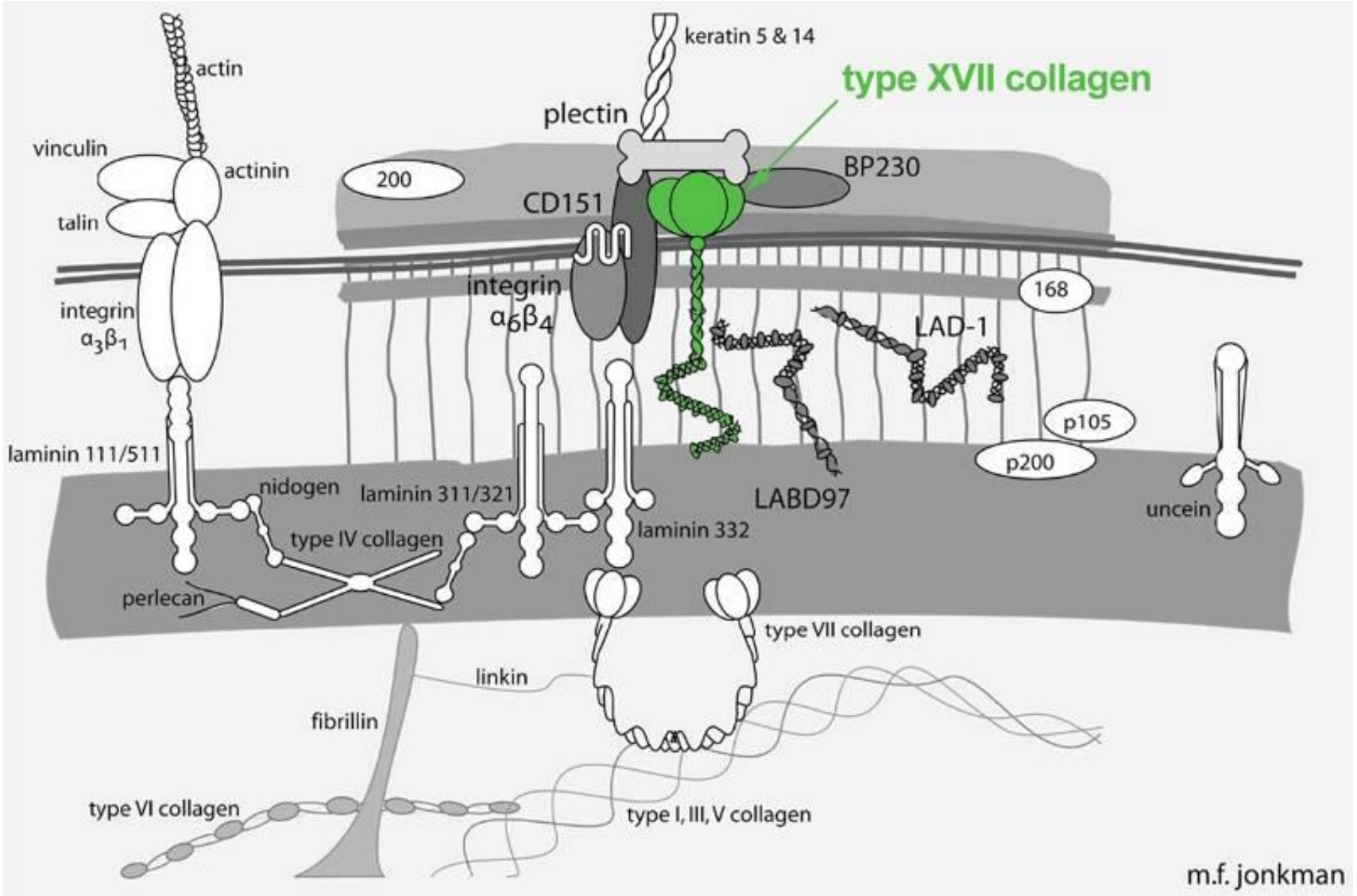
DEJ: Structural Organization





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Collagen XVII: Location At The Basement Membrane Zone And Hemidesmosome



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Syniorage™

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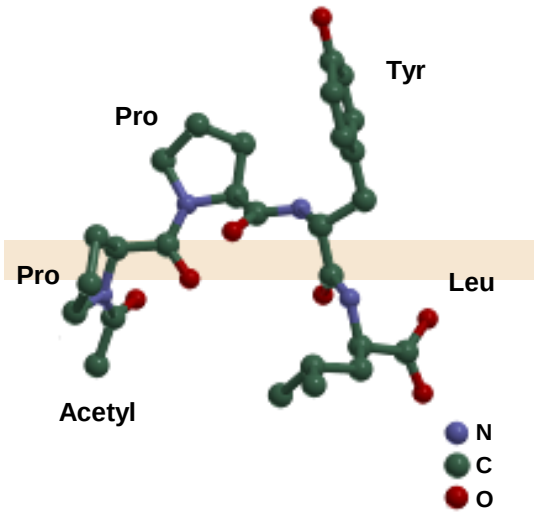
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Syniorage™: A White Powder Containing an Acetylated Tetrapeptide

Composition : Mannitol (and) Acetyl Tetrapeptide-11

Origin : synthetic

Patent application n° : WO 2006042661



PPYL motif : Sequence 138-141 of the antibacterial cathelicidin PR-39

Cathelicidin PR-39 stimulates syndecan expression

Gallo RL, Ono M, Povsic T, Page C, Eriksson E, Klagsbrun M, Bernfield M: Syndecans, cell surface heparin sulphate proteoglycans, are induced by a prolin-rich antimicrobial peptide from wounds. Proc Natl Acad Sci USA 91: 11035-11039, 1994

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Efficacy Tests

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General Gene Expression Profile

Influence on gene expression of human keratinocytes (*in vitro* test)

Protocol

Seeding of human epidermal keratinocytes (4 donors)
(38, 39, 42 and 45 years old)

↓

Incubation for 3-5 days at 37°C, CO_2 = 5%

↓

Introduction of the product

↓

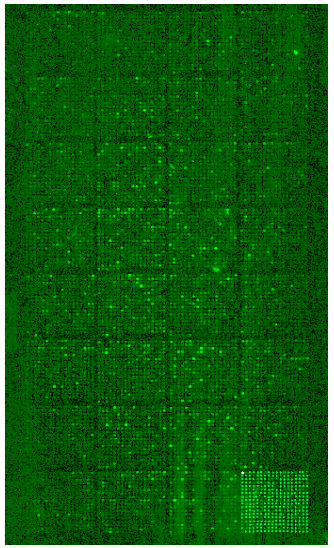
Control | Acetyl Tetrapeptide-11 at 0.87 µg/ml

↓

Incubation until 24 hours at 37°C, CO_2 = 5%

↓

Extraction of total RNAs
cDNA synthesis and labeling by fluorescent markers Cya3 or Cya5
Competitive hybridization on DNA-array slide

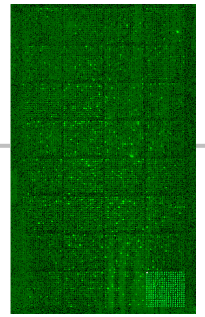


DNA array

(1300 genes)

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General Gene Expression Profile



DNA array

(1300 genes)

Influence on gene expression of human keratinocytes (*in vitro* test)

↘ expression of DDR1 receptor

↗ expression of COL17A1 gene

↗ cell growth

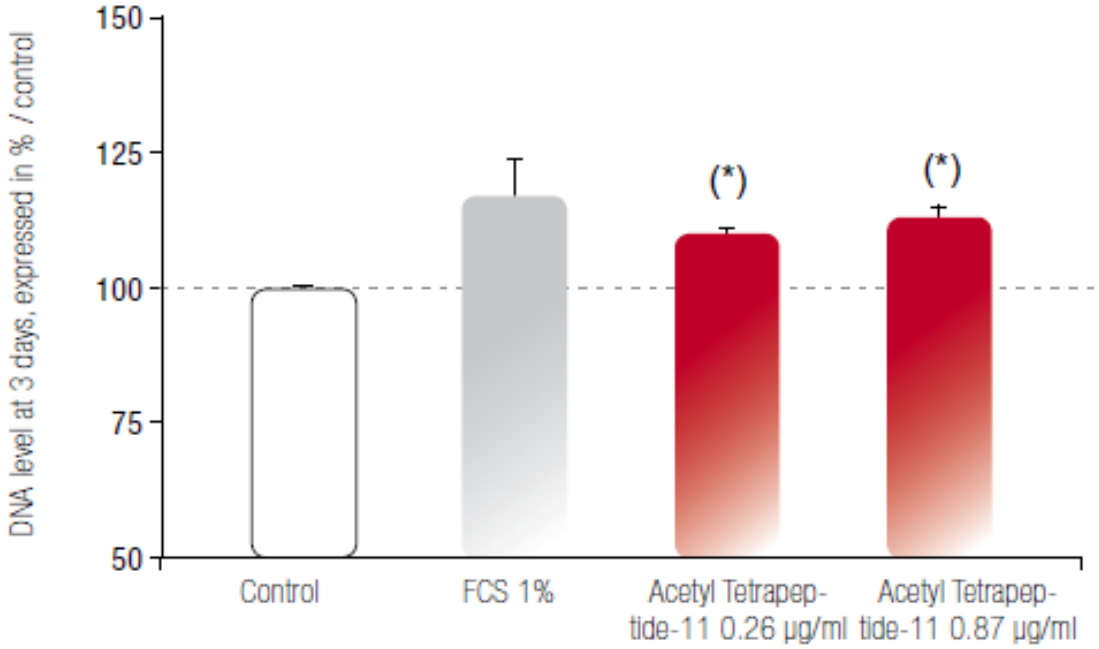
↗ synthesis of syndecan-1

DDR1: Discoidin Domain Receptor 1. The activation of this gene induces a decrease of cell growth and modulates expression of many genes of the ECM, including a decrease of syndecan-1 proteoglycan expression.

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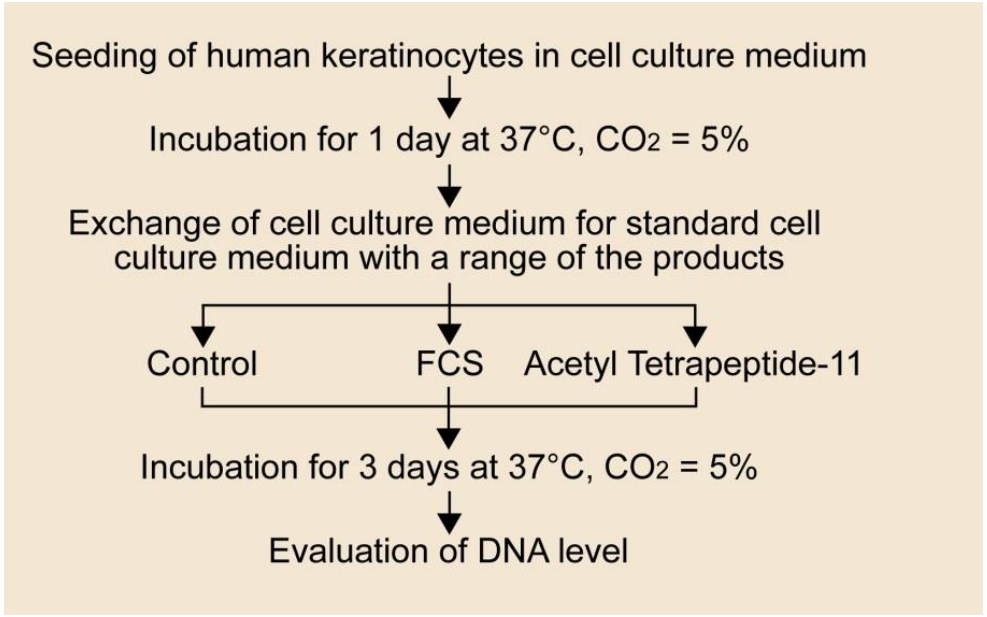
Effect On The Growth (Test On Human Keratinocytes)

Results



Mean ± SEM
3 assays in triplicate
Student's t test
(*) $p<0.05$

Protocol

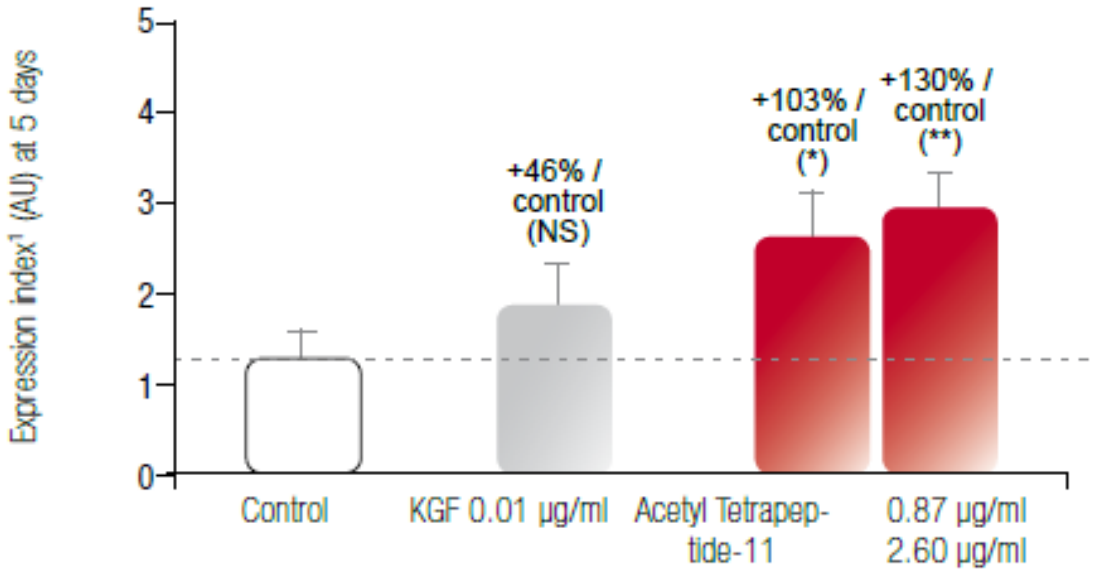


Conclusion

Acetyl Tetrapeptide-11 has significantly increased the growth of human keratinocytes (+13% at 0.87 µg/ml).

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Effect On Syndecan-1 Synthesis (Test On Human Keratinocytes)

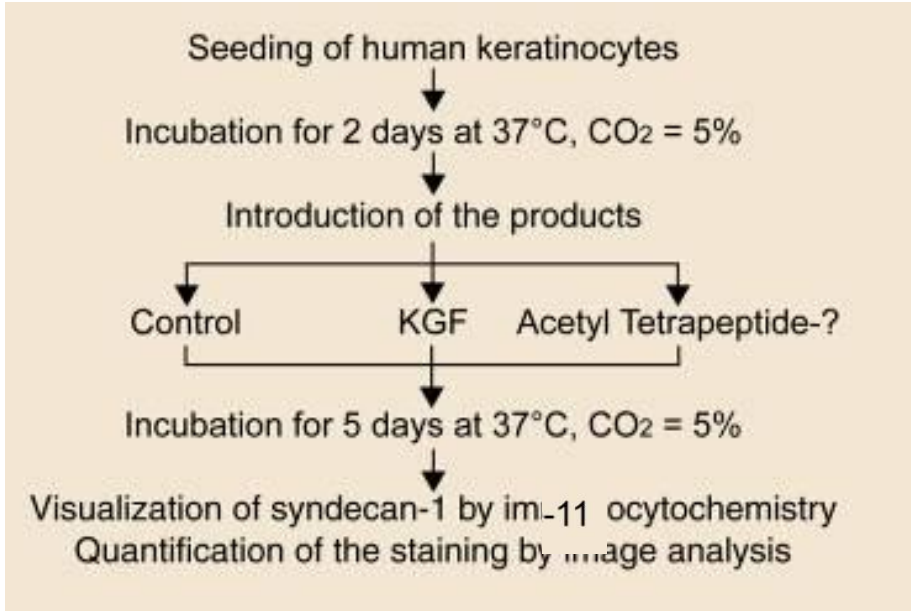


[1] Number of stained pixels x fluorescence intensity in green channel in arbitrary unit (AU)

- Stimulation of syndecan-1 synthesis by human keratinocytes: quantification.

Statistics:
Mean ± SEM
1 factor ANOVA with PLSD of Fisher
n=28 (4 assays)
(NS) Not significant
(*) $p<0.05$
(**) $p<0.01$

Protocol

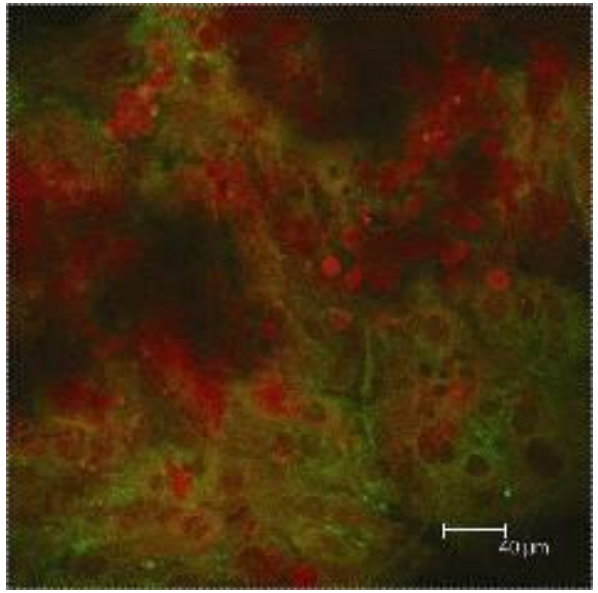


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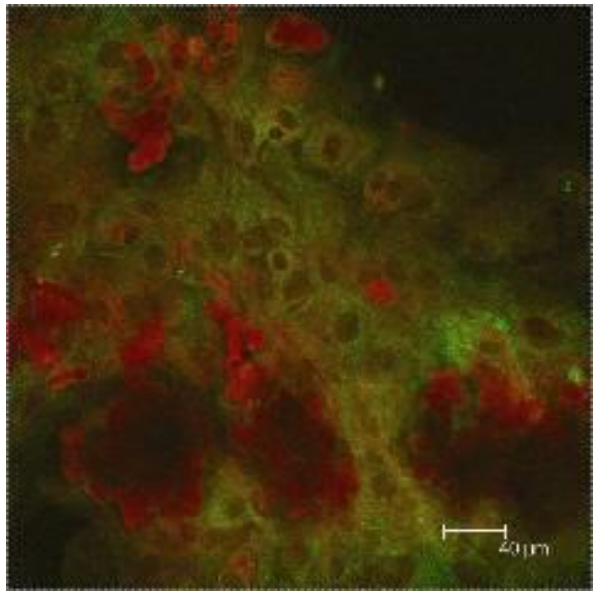
Effect On Syndecan-1 Synthesis (Test On Human Keratinocytes)

Results

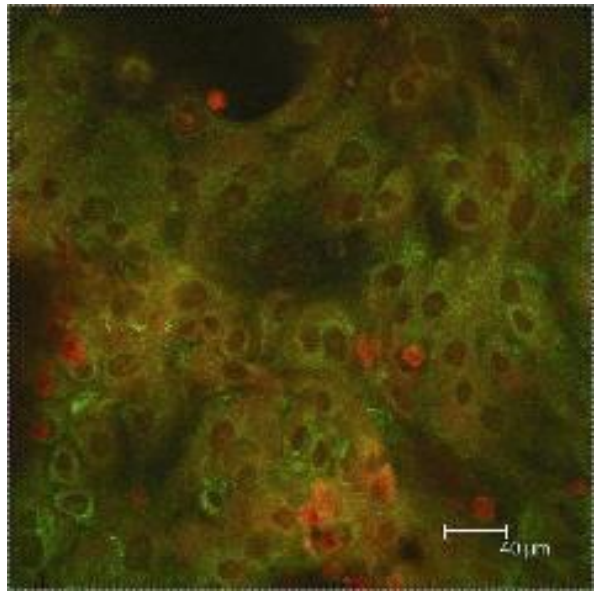
Control



KGF at 0.01 µg/ml



Acetyl Tetrapeptide-11
at 2.60 µg/ml

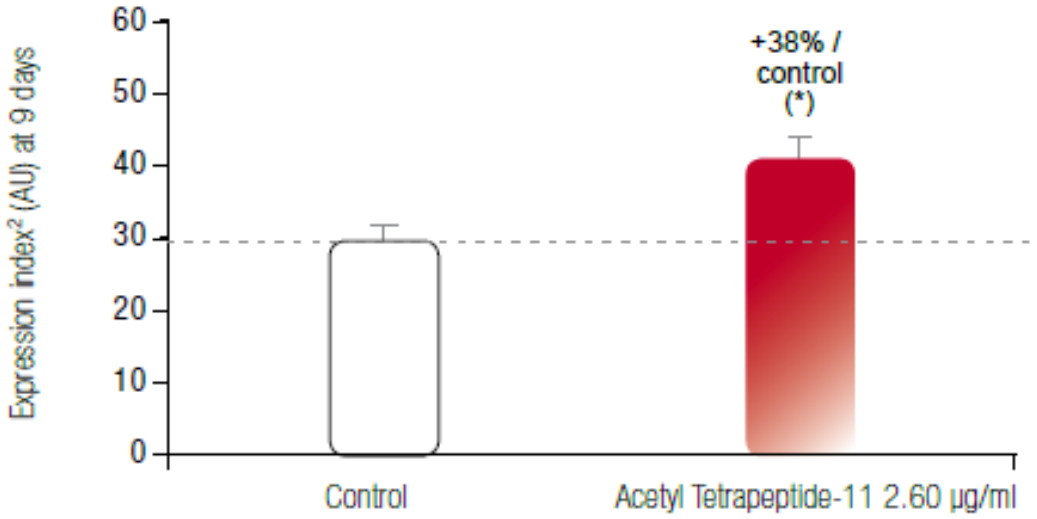


Conclusion

Acetyl Tetrapeptide-11 has significantly increased the synthesis of syndecan-1 by human keratinocytes in culture, in a dose dependant way. These results confirm one of the levels of action of Syniorage™: stimulation of a specific skin component, the small proteoglycan, syndecan-1.

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Effect On Syndecan-1 Synthesis



Mean ± SEM
1 factor ANOVA with PLSD of Fisher
n=8 (3 assays)
(*) $p<0.02$

² Number of stained pixels x fluorescence intensity in green channel/epidermal thickness in arbitrary unit (AU)

Stimulation of syndecan-1 synthesis by human reconstructed skin: quantification.

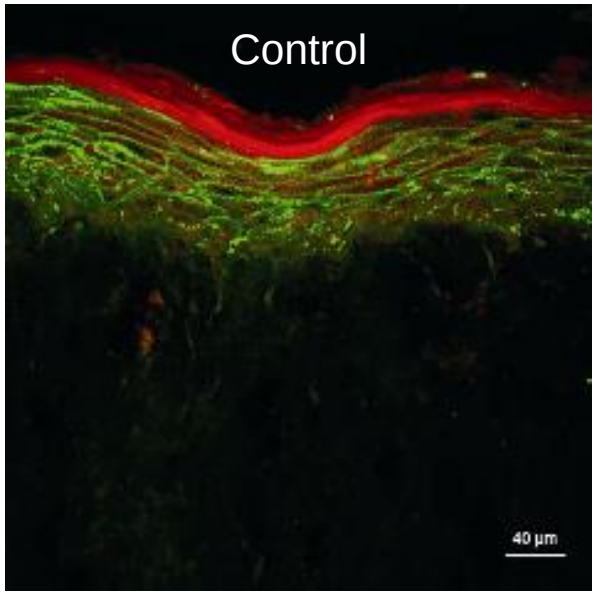
Protocol

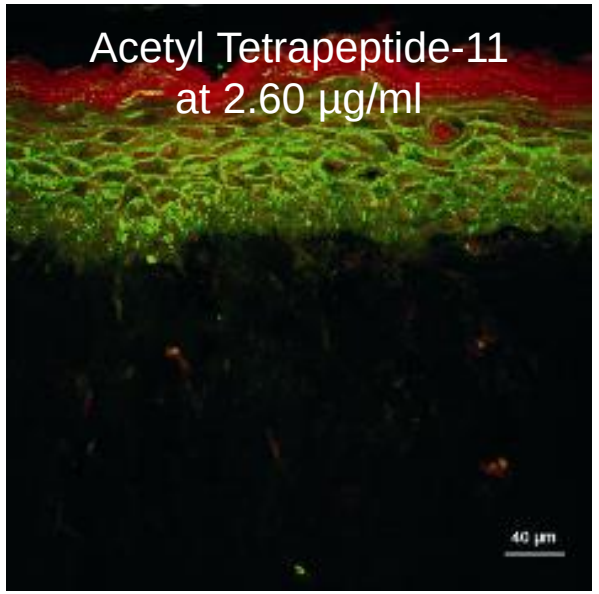
Preparation of human reconstructed skin
↓
Incubation for 1 day at 37°C, CO_2 = 5%
↓
Introduction of the product
↓
Control / Acetyl Tetrapeptide-11
↓
Incubation for 9 days at 37°C, CO_2 = 5%
↓
Biopsies
Visualization of syndecan-1 by immunohistochemistry
Quantification of the staining in the epidermis by image analysis

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Effect On Syndecan-1 Synthesis (Test On Full Thickness Skin Model)

Results





Conclusion

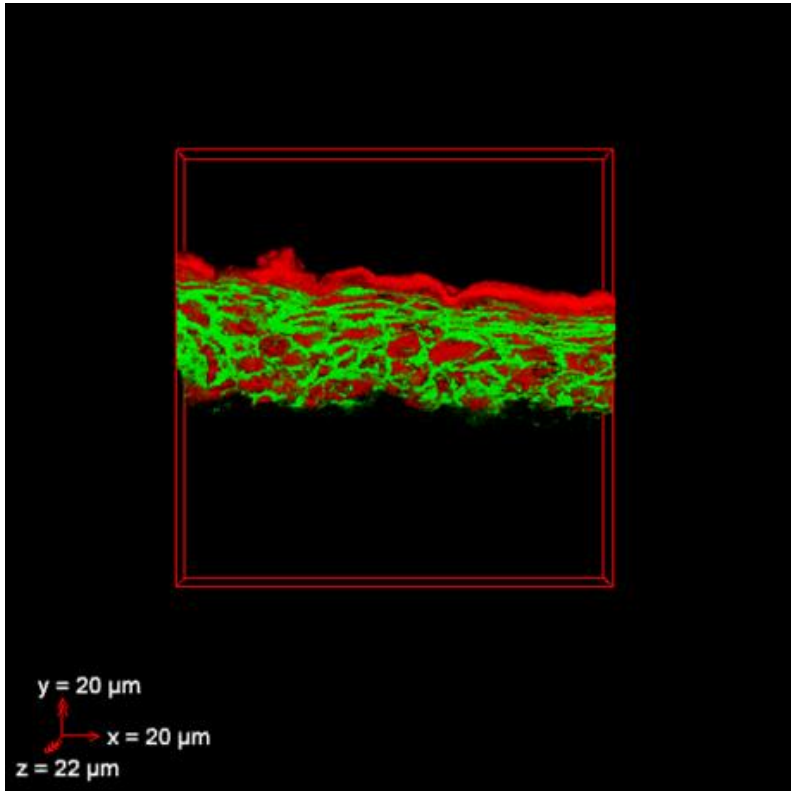
Acetyl Tetrapeptide-11 has clearly boosted the synthesis of syndecan-1 in this full thickness model, confirming the results obtained in the previous test on keratinocyte cultures and the targeted action of Syniorage™ on the small proteoglycan, syndecan-1.

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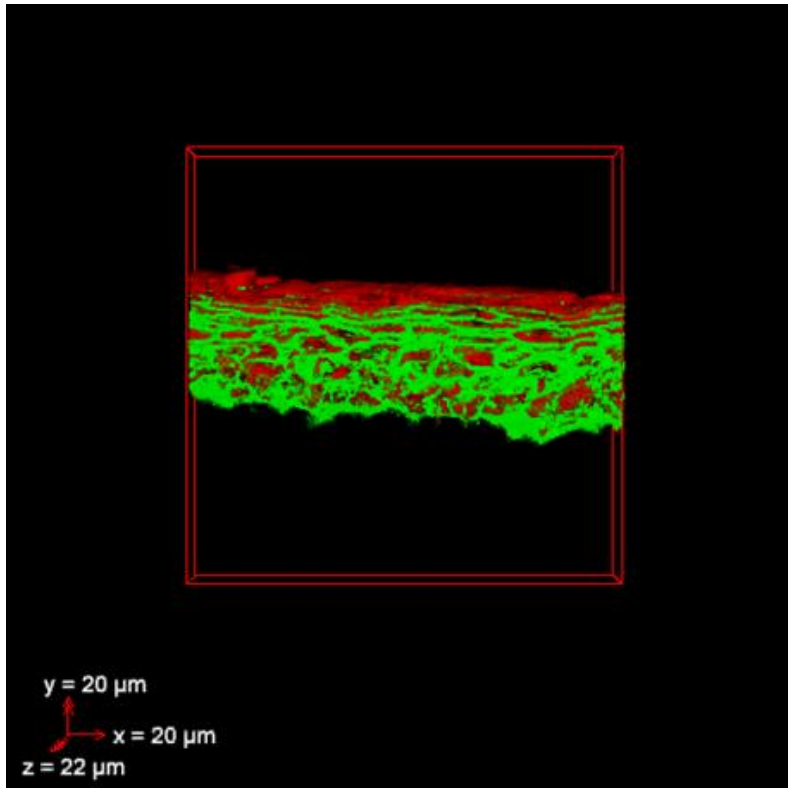
Effect On Syndecan-1 Synthesis (Test On Full Thickness Skin Model)

3D representation on 2 serial sections at 9 days

Control

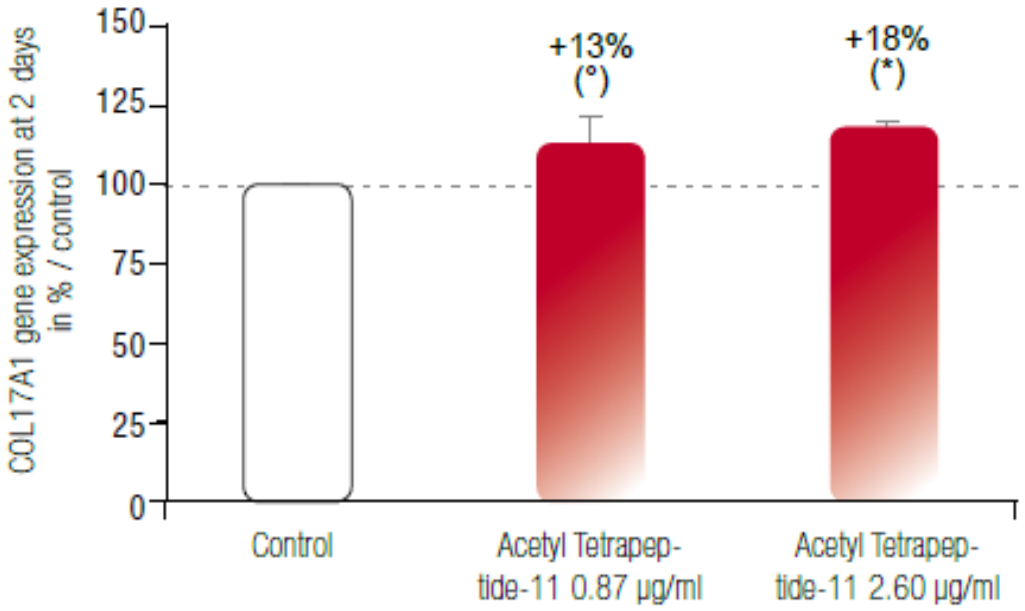


Acetyl Tetrapeptide-11 at 2.60 µg/ml



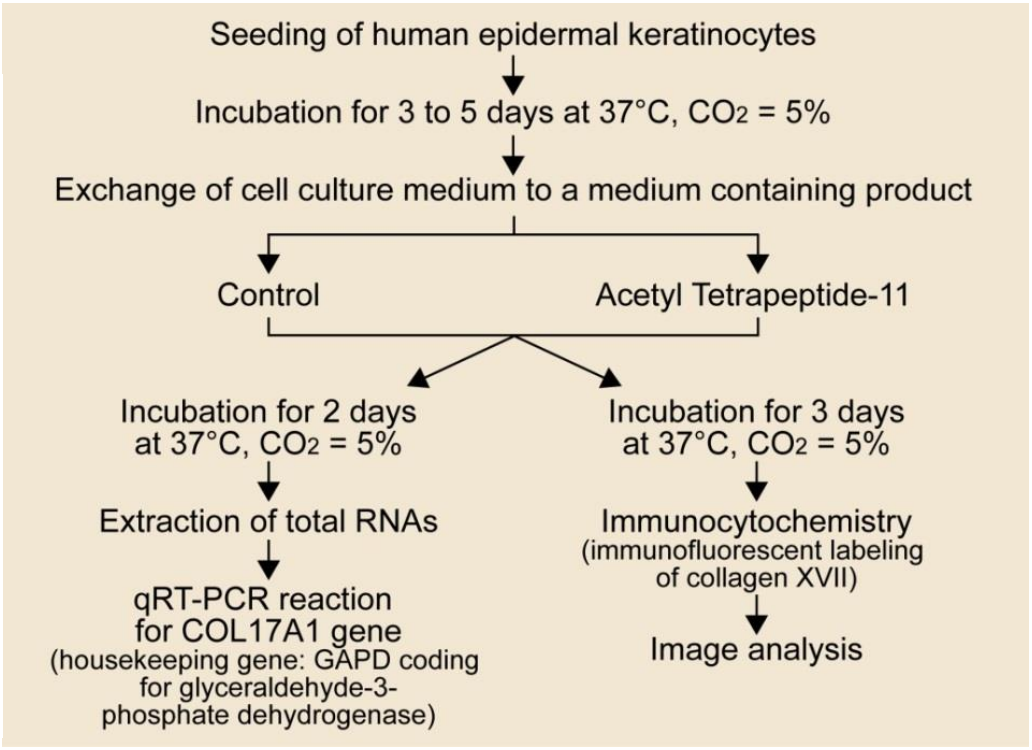
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Effect on the expression of collagen XVII gene
(*In vitro* test on human keratinocytes)



Mean ± SEM
n=4
Test U of Mann & Whitney / control
(°) p=0.08
(*) p<0.05

Protocol

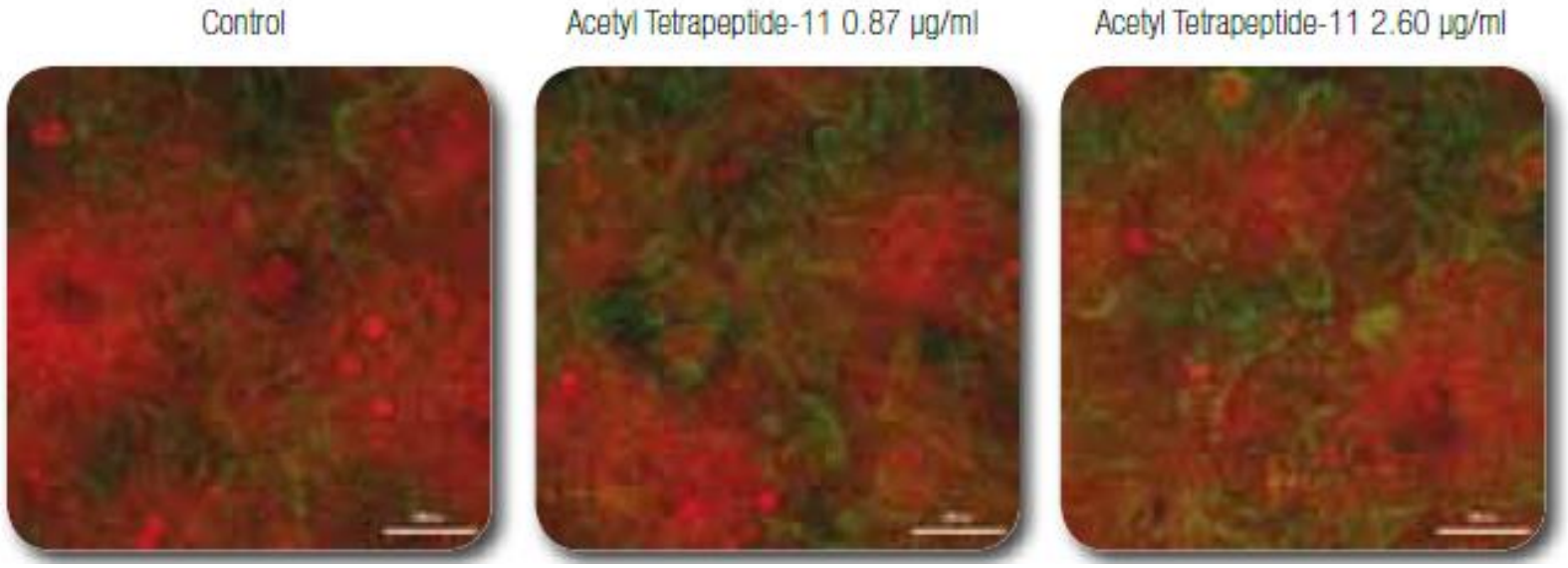


- Stimulation of the expression of collagen XVII COL17A1 gene by human epidermal keratinocytes: quantification.

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Effect on the expression of collagen XVII gene
(test on human keratinocytes)

Results



Stimulation of the synthesis of collagen XVII protein by human epidermal keratinocytes: visualization at 3 days.

Conclusion

Acetyl Tetrapeptide-11 has significantly increased the COL17A1 gene expression by human epidermal keratinocytes in culture. This result has been confirmed by visualizing the production of collagen XVII protein on cell culture.

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Anti-aging effect (clinical test)

Protocol

19 female volunteers, 60 to 70 years old,
with a loss of elasticity of the face

Bi-daily randomized application on the face for 56 days

Placebo cream



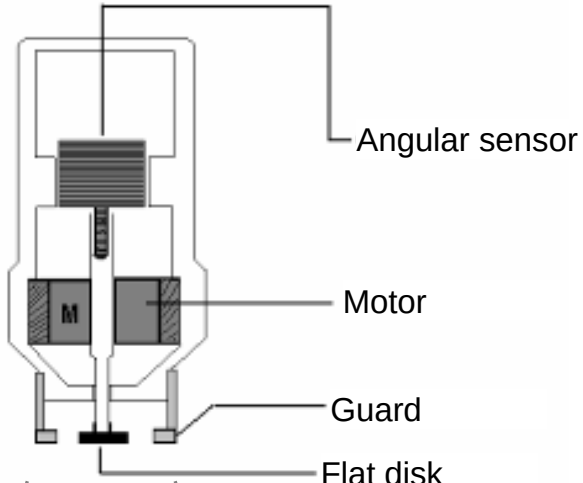
Cream with 3%
Syniorage™

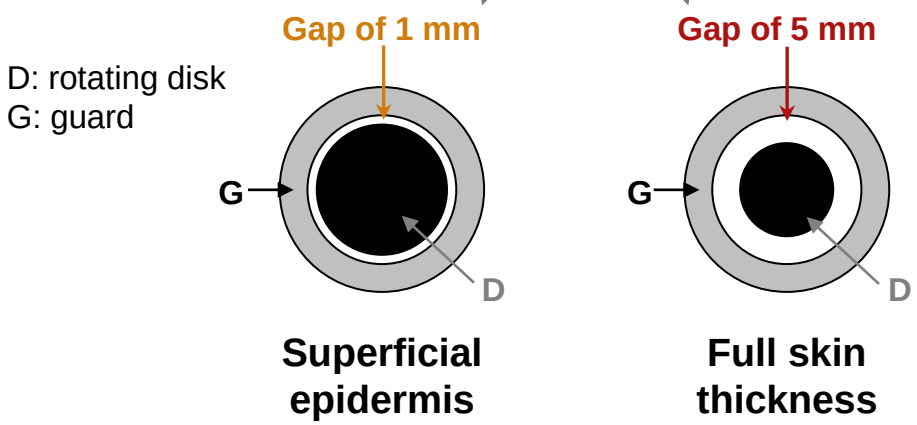
Macrophotographs of temples
Measurement of the biomechanical properties of the skin
on the upper cheek by torquemeter before treatment (D0),
and after 28 (D28) and 56 (D56) days of treatment

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Measurement of skin elasticity and firmness using a torquemeter

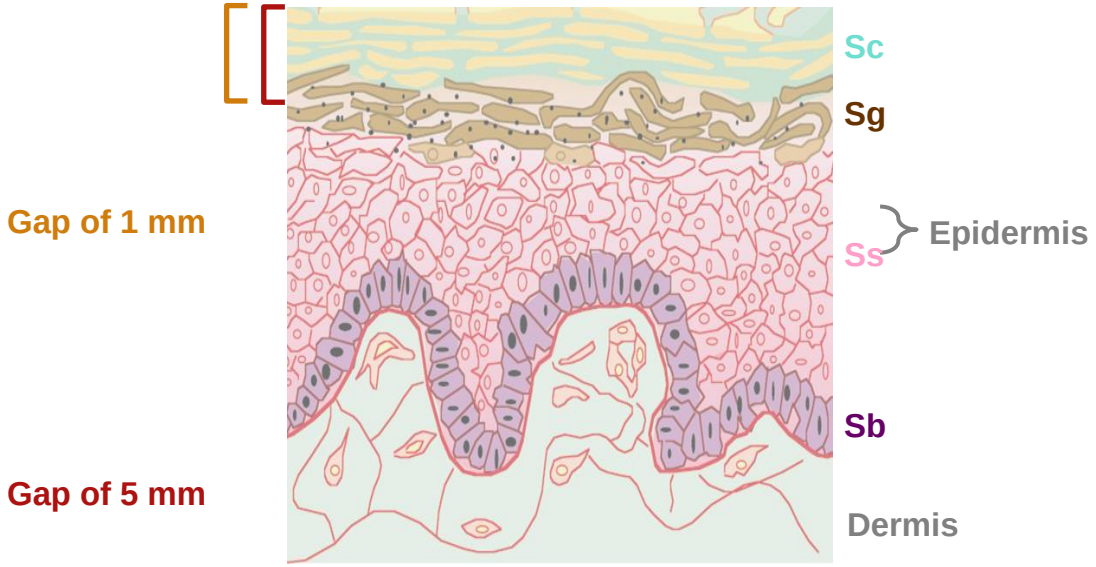
Principle of torquemeter





Measurement of angular displacement of skin in response to torsional forces applied by the torquemeter incorporated to the probe.

Gap of 1 mm characterizes the mechanical behaviour of **epidermis superficial layers**.

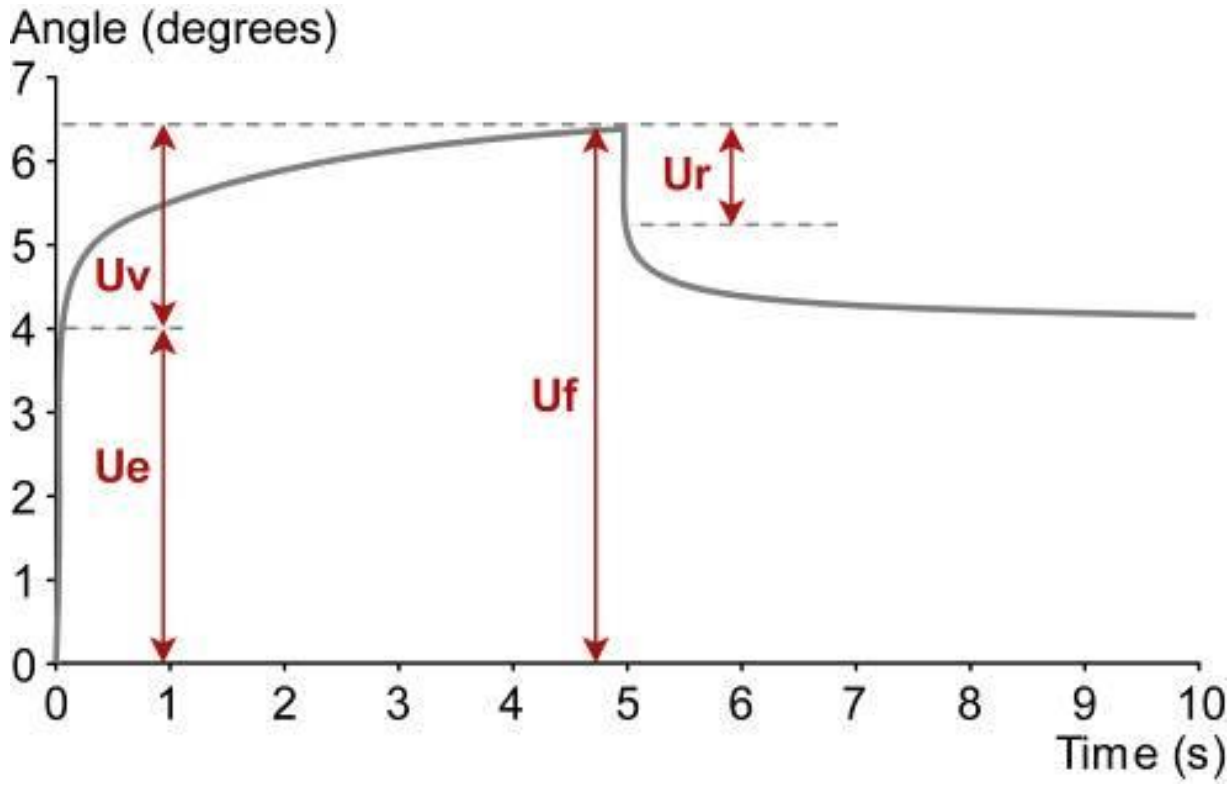


Agache PG: Twistometry measurement of skin elasticity. In: Non-invasive methods & skin. Serup J ed, Jemec GBE ed, 319-328, 1995

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Measurement of skin elasticity and firmness using a torquemeter

Measured parameters



$$\text{Skin elasticity} = \uparrow \frac{Ur}{Ue}$$

$$\text{Skin firmness} = \uparrow \frac{Ur}{Uf}$$

Ue: immediate extensibility

Uf: maximal extensibility

Uv: visco-elasticity

Ur: immediate elastic recovery

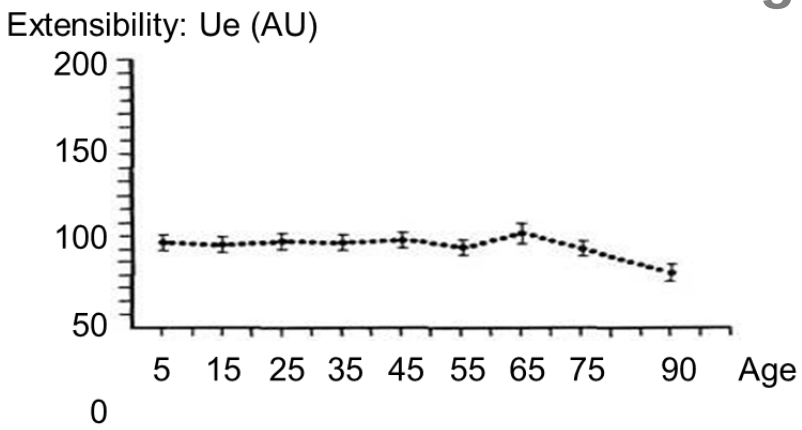
Skin deformation obtained with a torquemeter DTM 310

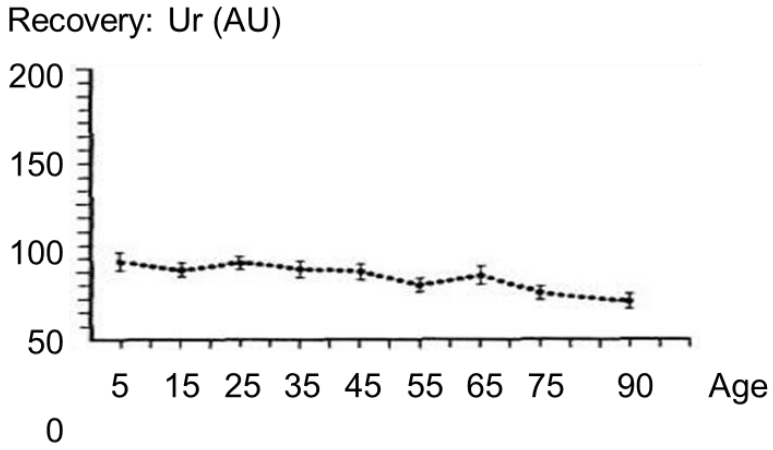
Agache PG: Twistometry measurement of skin elasticity. In: Non-invasive methods & skin. Serup J ed, Jemec GBE ed, 319-328, 1995

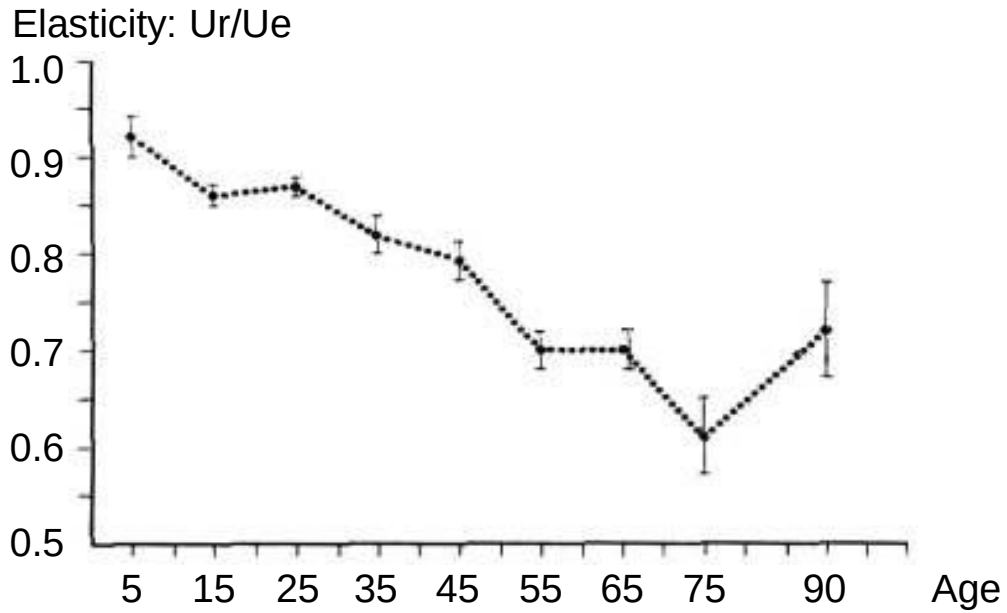
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Measurement of skin elasticity and firmness using a torquemeter

Influence of age on biomechanical properties of the skin







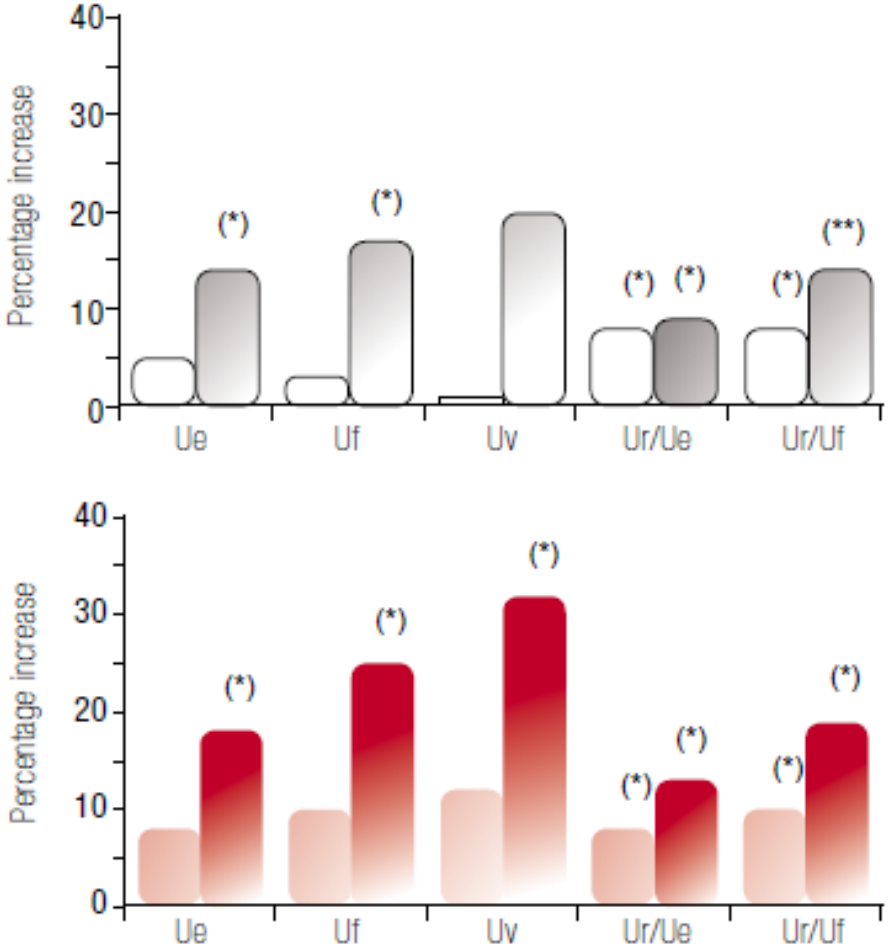
AU: arbitrary unit

Escoffier C, *et al.*: Age-related mechanical properties of human skin: an *in vivo* study. J Invest Dermatol, 93(3):353-357, 1989

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Measurement of skin elasticity and firmness on the upper cheek using a torquemeter (on volunteers, 60 to 70 years old)

Results



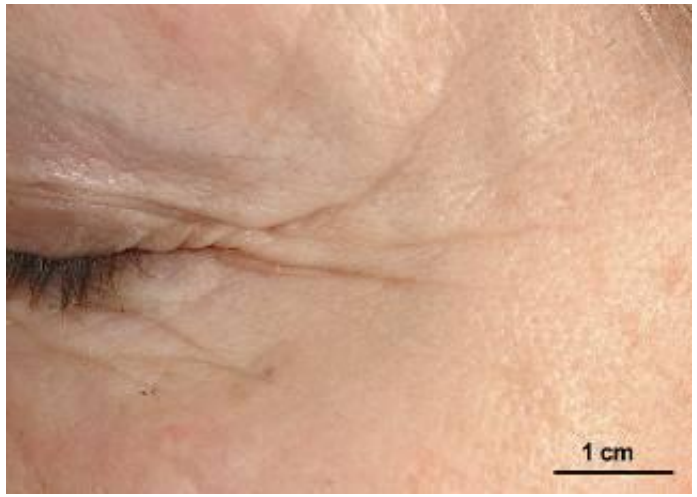
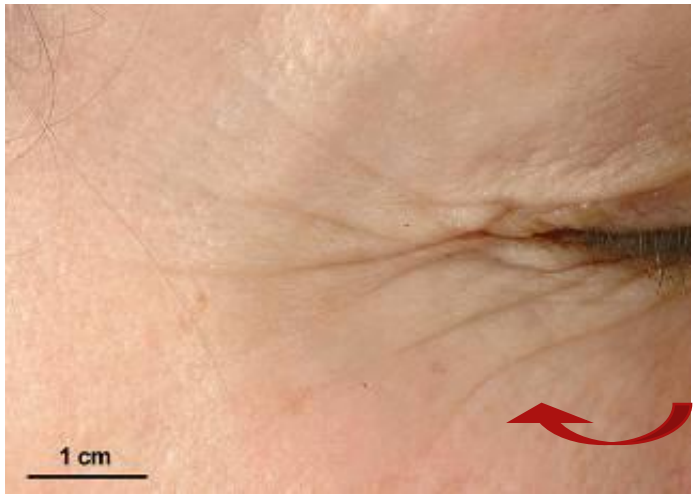
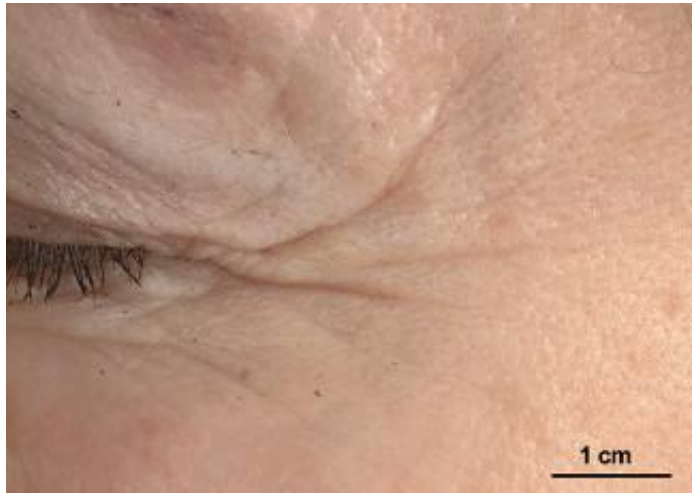
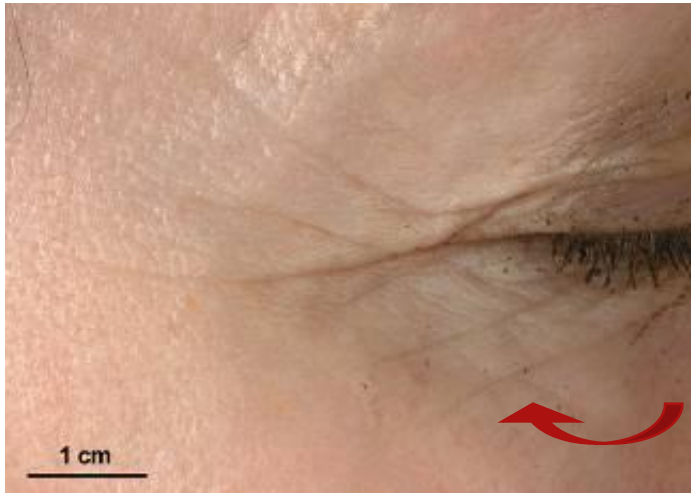
Percentage increase
of 19 volunteers
Paired Student's t test / D0
(*) $p<0.05$
(**) $p<0.001$

D0: before treatment
D28: after 28 days of treatment
D56: after 56 days of treatment

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Anti-aging effect (clinical test)

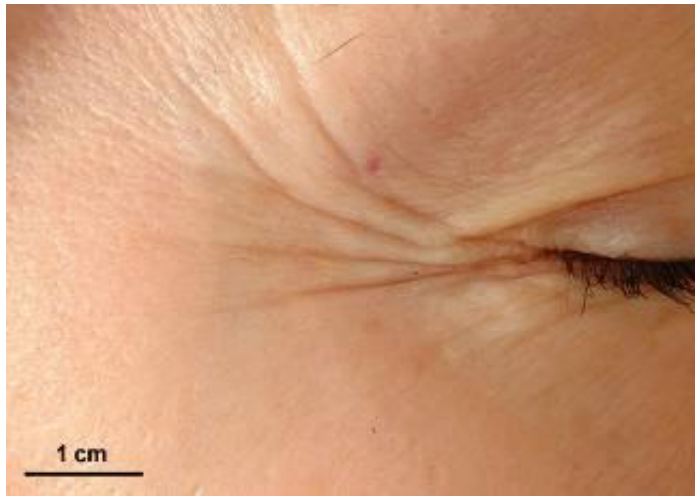
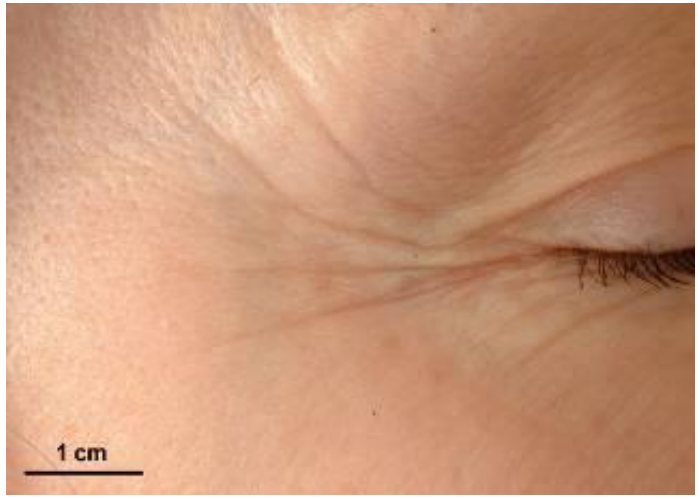
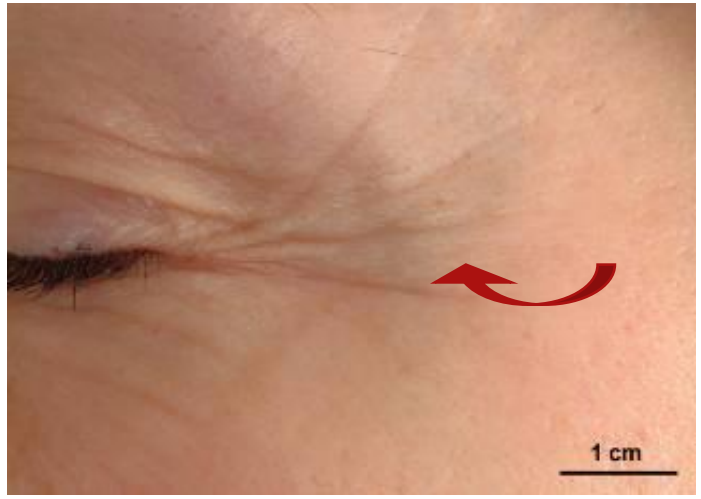
Results

	Placebo cream	Cream with 3% Syniorage™
Before treatment (D0)		
After 56 days of treatment (D56)		

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Anti-aging effect (clinical test)

Results

	Placebo cream	Cream with 3% Syniorage™
Before treatment (D0)		
After 56 days of treatment (D56)		

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Syniorage™

Conclusion

The significant increase of the biomechanical parameters of the superficial layers of the epidermis, *i.e.* firmness and elasticity, reflects an improvement of epidermal cohesion and consequently an anti-aging effect.

The cream with 1% SYNIORAGE™ has a 5 - 10% better effect than the placebo cream.

Skin texture was improved, skin relief was smoothed and skin radiance was increased after 56 days of treatment.

By increasing the production of syndecan-1, SYNIORAGE™ has allowed to increase skin elasticity and firmness, and thus to improve skin resistance. By stimulating the production of collagen XVII, SYNIORAGE™ allows to improve the cohesion of the different skin layers, for more uniform and thus more radiant skin.

Syniorage™

Influence on gene expression of human keratinocytes (*in vitro* test)

DNA array

↘ expression of DDR1 receptor

↗ cell growth

↗ synthesis of syndecan-1

⇗ of cohesion between epidermal upper layers

↗ expression of COL17A1 gene

↗ COL17A1 gene

↗ collagen XVII protein

⇗ of cohesion between epidermis and dermis (hemidesmosomes)

Visible and measurable results on epidermis superficial layers in vivo

⇗ skin appearance and firmness

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Skin Benefits

There is a discrepancy between the spirit of seniors and the appearance of their skin.

Whereas with age they acquire strength and serenity, their skin on the contrary becomes more fragile and blemished.

By targeting 2 specific constituents of the epidermis responsible for its cohesion, syndecan-1 and collagen XVII, Syniorage™ helps to reestablish the parallel between the life of seniors and the appearance of their skin: more resistant and radiant, it reflects the wisdom acquired by experience and their joy in life.

For an optimal approach, it is recommended to use SYNIORAGE™, specific for the epidermis, in conjunction with DERMICAN™, which specifically targets aging of the dermis.

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Cosmetic Applications

Anti-age face care

Specific care for mature skin

Radiant senior skin care



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Technical Data

SYNIORAGE™ PW LS 9847

Aspect	:	white powder with a weak characteristic odor
Dose of use	:	0.5 – 1.5%
Solubility	:	Soluble in water, insoluble in oils and fats
INCI Name	:	Mannitol (and) Acetyl Tetrapeptide-11
Preservatives	:	none

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GOLDEN AGE
Specific care for mature skin

Ingredients	INCI	%	Function
Phase I			
Emulgade Sucro	**Sucrose Polystearate (and) Hydrogenated Polyisobutene**	**2.0**	**O/W cream base SE**
Cutina PES	**Pentaerythrityl Distearate**	**2.0**	**Consistency giving factor**
Cetiol 868	**Ethylhexyl Stearate**	**3.0**	**Emollient**
Myritol 318	**Caprylic/Capric Triglyceride**	**2.0**	**Emollient**
Cetiol CC	**Dicaprylyl Carbonate**	**2.0**	**Emollient**
Cetiol PGL	**Hexyldecanol (and) Hexyldecyl Laurate**	**3.0**	**Emollient**
Cosmedia SP	**Sodium Polyacrylate**	**0.6**	**Thickener**
DC 200-50 cts (1)	Dimethicone	1.0	Defoaming agent
Phase II			
Water	Aqua	75.45	
Glycerin	Glycerin	3.0	Humectant
Elestab 388	**Propylene glycol (and) Phenoxyethanol (and) Chlorphenesin (and) Methylparaben**	**2.5**	**Preservative**
Keltrol CGT (2)	Xanthan Gum	0.1	Thickener
Eumulgin SG	**Sodium Stearoyl Glutamate**	**0.3**	**Emulsifier**
Phase III			
Syniorage LS 9748	**Water (and) Glycerin (and) Acetyl Tetrapeptide -11**	**3.0**	**Active**
Fragrance	Fragrance	0.05	

Suppliers
(1) Dow Corning
(2) Kelco

PROCEDURE:
Heat the oil phase I to 75-80°C and mix homogeneously. Heat phase II to 75-80°C and add to the oil phase while stirring. Allow the emulsion to cool while stirring in such a way that it remains in continual motion. Avoid incorporation of air. If necessary homogenize with a suitable dispersion unit (e.g. Ultra Turrax) approx. 55°C. Add phase III below 40°C with mixing. Stir while cooling until room temperature.

Viscosity: (Brookfield RVT, 23C, spindle#5, 10rpm): *27,000 cps*
pH value: *6.9*

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